

NCERT Solutions for Class 12 Mathematics

Chapter 9 – Differential Equations

Exercise 9.1

1. Determine order and degree (if defined) of differential equation $\frac{d^4y}{dx^4} + \sin(y''') = 0$

Ans: Rewrite the equation $\frac{d^4y}{dx^4} + \sin(y''') = 0$ as:

$$\Rightarrow y'''' + \sin(y''') = 0$$

The highest order between the two terms is of y'''' which is four.

The differential equation contains a trigonometric derivative term and is not completely polynomial in its derivative, thus degree is not defined.

2. Determine order and degree (if defined) of differential equation $y' + 5y = 0$.

Ans: The given differential equation is $y' + 5y = 0$.

The highest order term is y' , thus the order is one.

As the derivative is of completely polynomial nature is and highest power of derivative is of y' which is one. Thus, degree is one.

3. Determine order and degree (if defined) of differential equation $\left(\frac{ds}{dt}\right)^4 + 3s \frac{d^2s}{dt^2} = 0$.

Ans: The given differential equation is $\left(\frac{ds}{dt}\right)^4 + 3s \frac{d^2s}{dt^2} = 0$.

The highest order term is $\frac{d^2s}{dt^2}$, thus the order is two.

As the derivative is of completely polynomial nature is and highest power of derivative term $\frac{d^2s}{dt^2}$ which is one. Thus, the degree is one.

4. Determine order and degree (if defined) of differential equation $\left(\frac{d^2y}{dx^2}\right)^2 + \cos\left(\frac{dy}{dx}\right) = 0$.

Ans: The given differential equation is $\left(\frac{d^2y}{dx^2}\right)^2 + \cos\left(\frac{dy}{dx}\right) = 0$.

The highest order term is $\frac{d^2y}{dx^2}$, thus the order is two.

The differential equation contains a trigonometric derivative term and is not completely polynomial in its derivative, thus degree is not defined.

5. Determine order and degree (if defined) of differential equation $\left(\frac{d^2y}{dx^2}\right) - \cos 3x + \sin 3x = 0$.

Ans: The given differential equation is $\left(\frac{d^2y}{dx^2}\right) - \cos 3x + \sin 3x = 0$.

The highest order term is $\frac{d^2y}{dx^2}$, thus the order is two.

As the derivative is of completely polynomial nature is and highest power of derivative term $\frac{d^2y}{dx^2}$ which is one. Thus, degree is one.

6. Determine order and degree (if defined) of differential equation $(y''')^2 + (y'')^3 + (y')^4 + y^5 = 0$.

Ans: The given differential equation is $(y''')^2 + (y'')^3 + (y')^4 + y^5 = 0$.

The highest order term is $(y''')^2$, thus the order is three.

The differential equation is of the polynomial form and the power of highest order term y''' is two, thus the degree is two.

7. Determine order and degree (if defined) of differential equation $y''' + 2y'' + y' = 0$.

Ans: The given differential equation is $y''' + 2y'' + y' = 0$.

The highest order derivative in the differential equation is y''' . Thus, its order is three.

The differential equation is polynomial with the highest order term y''' having a degree one. Thus, the degree is one.

8. Determine order and degree (if defined) of differential equation $y' + y = e^x$.

Ans: The given differential equation is $y' + y = e^x$

Therefore:

$$\Rightarrow y' + y - e^x = 0$$

The highest order derivative in the differential equation is y' . Thus, its order is one.

The given equation is of polynomial form with the highest order term y' with degree one. Thus, the degree is one.

9. Determine order and degree (if defined) of differential equation $y'' + (y')^2 + 2y = 0$.

Ans: The given differential equation is $y'' + (y')^2 + 2y = 0$.

The highest order derivative in the differential equation is y'' . Thus, its order is two.

The given equation is of polynomial form with the highest order term y'' with highest degree one. Thus, the degree is one.

10. Determine order and degree (if defined) of differential equation $y'' + 2y' + \sin y = 0$.

Ans: The given differential equation is $y'' + 2y' + \sin y = 0$.

The highest order derivative in the differential equation is y'' . Thus, its order is two.

The given equation is of polynomial form with the highest order term y'' with the highest degree one. Thus, the degree is one.

11. Find the degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + \sin\left(\frac{dy}{dx}\right) + 1 = 0$.

(A) 3

(B) 2

(C) 1

(D) not defined

Ans: The given differential equation is $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + \sin\left(\frac{dy}{dx}\right) + 1 = 0$.

The differential equation is not polynomial in its derivative because of the term $\sin\left(\frac{dy}{dx}\right)$ thus its order is not defined.

The correct answer is (D).

12. Find the order of the differential equation $2x^2 \frac{d^2y}{dx^2} - 3 \frac{dy}{dx} + y = 0$.

(A) 2

(B) 1

(C) 0

(D) not defined

Ans: The given differential equation is $2x^2 \frac{d^2y}{dx^2} - 3 \frac{dy}{dx} + y = 0$.

The highest order term of the equation is $\frac{d^2y}{dx^2}$, thus the order is two.

The correct answer is (A).

Exercise 9.2

In each of the Questions, 1 to 10 verifies that the given functions (explicit or implicit) is a solution of the corresponding differential equation:

1. $y = e^x + 1 : y'' - y' = 0$.

Ans: The given equation is $y = e^x + 1$.

Differentiating equation on both sides for x , the equation becomes,

$$\frac{dy}{dx} = \frac{d}{dx}(e^x + 1)$$

The derivative of e^x is e^x and the derivative of a constant, $\frac{d}{dx}(a) = 1$, where a is constant.

Thus,

$$\Rightarrow \frac{dy}{dx} = e^x$$

$$\Rightarrow y' = e^x$$

Now, differentiate the obtained equation again for x ,

$$\Rightarrow \frac{d}{dx}(y') = \frac{d}{dx}(e^x)$$

$$\Rightarrow y'' = e^x$$

Therefore, consider the given equation to be considered for evaluation, $y'' - y' = 0$.

Now, the left-hand side of the given equation is $y'' - y'$.

Substitute e^x for y'' and e^x for y' in the left-hand side of the equation,

$$\Rightarrow y'' - y' = e^x - e^x$$

$$\Rightarrow y'' - y' = 0$$

Therefore, the LHS of the equation is found to be equivalent to the right hand side of the equation.

As a result, the provided function is the differential equation's solution.

$$2. y = x^2 + 2x + C : y' - 2x - 2 = 0.$$

Ans: The given equation is $y = x^2 + 2x + C$.

Differentiating equation on both sides with respect to x , the equation becomes,

$$y' = \frac{d}{dx}(x^2 + 2x + C)$$

The derivative of x^n is nx^{n-1} and the derivative of x , $\frac{d}{dx}(x) = 1$, whereas the derivative $\frac{d}{dx}(a) = 1$, where a is constant.

Thus,

$$\Rightarrow y' = \frac{d}{dx}(x^2 + 2x + C)$$

$$\Rightarrow y' = \frac{d}{dx}(x^2) + \frac{d}{dx}(2x) + \frac{d}{dx}(C)$$

$$\Rightarrow y' = 2x + 2$$

Now, consider the LHS of the equation that is to be taken for evaluation, $y' - 2x - 2 = 0$.

Substitute the obtained value for y' in the LHS of the above equation.

$$\Rightarrow y' - 2x - 2 = (2x - 2) - 2x - 2$$

$$\Rightarrow y' - 2x - 2 = 2x - 2 - 2x + 2$$

$$\Rightarrow y' - 2x - 2 = 0$$

Therefore, the LHS of the equation is found to be equivalent to the right hand side of the equation.

As a result, the provided function is the differential equation's solution.

3. $y = \cos x + C : y' + \sin x = 0$.

Ans: The given equation is $y = \cos x + C$.

Differentiating equation on both sides with respect to x , the equation becomes,

$$y' = \frac{d}{dx}(\cos x + C)$$

The derivative of $\cos x$ is $-\sin x$, whereas the derivative $\frac{d}{dx}(a) = 1$, where a is constant.

Thus,

$$\Rightarrow y' = \frac{d}{dx}(\cos x + C)$$

$$\Rightarrow y' = \frac{d}{dx}(\cos x) + \frac{d}{dx}(C)$$

$$\Rightarrow y' = -\sin x$$

Now, consider the LHS of the equation that is to be considered for evaluation, $y' + \sin x = 0$.

Substitute the obtained value for y' in the LHS of the above equation.

$$\Rightarrow y' + \sin x = (-\sin x) + \sin x$$

$$\Rightarrow y' + \sin x = 0$$

Therefore, the LHS of the equation is found to be equivalent to the right hand side of the equation.

As a result, the provided function is the differential equation's solution.

4. $y = \sqrt{1+x^2} : y' = \frac{xy}{1+x^2}$.

Ans: The given equation is $y = \sqrt{1+x^2}$.

Differentiating equation on both the sides of with respect to x , the equation becomes,

$$y' = \frac{d}{dx}(\sqrt{1+x^2})$$

The derivative of $\sqrt{x}$ is $\frac{1}{2\sqrt{x}}$ and the derivative of x^n , $\frac{d}{dx}(x^n) = nx^{n-1}$, whereas the derivative

$\frac{d}{dx}(a) = 1$, where a is constant.

Thus,

$$\Rightarrow y' = \frac{d}{dx}(\sqrt{1+x^2})$$

$$\Rightarrow y' = \frac{1}{2\sqrt{1+x^2}} \cdot \frac{d}{dx}(1+x^2)$$

$$\Rightarrow y' = \frac{2x}{2\sqrt{1+x^2}}$$

$$\Rightarrow y' = \frac{x}{\sqrt{1+x^2}}$$

Now, multiply and divide the numerator and denominator of the right-hand side of the above equation by $\sqrt{1+x^2}$.

$$\Rightarrow y' = \frac{x}{\sqrt{1+x^2}} \left(\frac{\sqrt{1+x^2}}{\sqrt{1+x^2}} \right)$$

$$\Rightarrow y' = \frac{x}{1+x^2} \cdot \sqrt{1+x^2}$$

Substitute y for $\sqrt{1+x^2}$ in the above equation,

$$\Rightarrow y' = \frac{x}{1+x^2} \cdot y$$

$$\Rightarrow y' = \frac{xy}{1+x^2}$$

Therefore, the LHS of the equation is found to be equivalent to the right hand side of the equation.

As a result, the provided function is the differential equation's solution.

5. $y = Ax : xy' = y (x \neq 0)$.

Ans: The given equation is $y = Ax$.

Differentiating equation on both the sides of with respect to x , the equation becomes,

$$y' = \frac{d}{dx}(Ax)$$

The derivative of x is equal to 1.

Thus,

$$\Rightarrow y' = \frac{d}{dx}(Ax)$$

$$\Rightarrow y' = A \frac{d}{dx}(x)$$

$$\Rightarrow y' = A(1)$$

$$\Rightarrow y' = A$$

Now, consider the LHS of the equation to be evaluated, that is, $xy' = y(x \neq 0)$.

Substitute the obtained value for y' in the LHS of the above equation.

$$\Rightarrow xy' = xA$$

$$\Rightarrow xy' = Ax$$

Thus, $y = Ax$.

Therefore, the LHS of the equation is found to be equivalent to the right hand side of the equation.

As a result, the provided function is the differential equation's solution.

6. $y = x \sin x : xy' = y + x\sqrt{x^2 - y^2} \quad (x \neq 0 \text{ and } x > y \text{ or } x < -y).$

Ans: The given equation is $y = x \sin x$.

Differentiating equation on both the sides with respect to x , the equation becomes,

$$y' = \frac{d}{dx}(x \sin x)$$

The product rule of differential equations says that,

$$\frac{d}{dx}(uv) = u \frac{d}{dx}(v) + v \frac{d}{dx}(u)$$

The derivative of $\sin x$ is $\cos x$ and the derivative of x is 1.

$$\Rightarrow y' = \frac{d}{dx}(x \sin x)$$

$$\Rightarrow y' = x \frac{d}{dx}(\sin x) + \sin x \frac{d}{dx}(x)$$

$$\Rightarrow y' = x \cos x + \sin x$$

Now, consider the LHS of the equation $xy' = y + x\sqrt{x^2 - y^2}$.

Substitute the obtained value for y' in the LHS of the above equation.

$$\Rightarrow xy' = x(x \cos x + \sin x)$$

$$\Rightarrow xy' = x^2 \cos x + x \sin x$$

It is known that $\sin^2 x + \cos^2 x = 1$, thus, $\cos x = \sqrt{1 - \sin^2 x}$ and $\sin x = \frac{y}{x}$ as $y = x \sin x$.

Thus,

$$\Rightarrow xy' = x^2 \sqrt{1 - \sin^2 x} + y$$

$$\Rightarrow xy' = x^2 \sqrt{1 - \left(\frac{y}{x}\right)^2} + y$$

$$\Rightarrow xy' = \frac{x^2}{x} \sqrt{x^2 - y^2} + y$$

$$\Rightarrow xy' = x \sqrt{x^2 - y^2} + y$$

$$\text{Thus, } xy' = y + x \sqrt{x^2 - y^2}$$

Therefore, the LHS of the equation is found to be equivalent to the right hand side of the equation.

As a result, the provided function is the differential equation's solution.

$$7. \text{ } xy = \log y + C : y' = \frac{y^2}{1 - xy} (xy \neq 1).$$

Ans: The given equation is $xy = \log y + C$.

Differentiating equation on both the sides with respect to x , the equation becomes,

$$\frac{d}{dx}(xy) = \frac{d}{dx}(\log y)$$

The product rule of differential equations says that,

$$\frac{d}{dx}(uv) = u \frac{d}{dx}(v) + v \frac{d}{dx}(u)$$

Also, the derivative of x is 1, whereas the derivative of $\log x$ is $\frac{1}{x}$.

Thus,

$$\Rightarrow \frac{d}{dx}(xy) = \frac{d}{dx}(\log y)$$

$$\Rightarrow y \frac{d}{dx}(x) + x \frac{d}{dx}(y) = \frac{d}{dx}(\log y)$$

$$\Rightarrow y(1) + xy' = \frac{1}{y} y'$$

$$\Rightarrow y^2 + xyy' = y'$$

Now, consider the LHS of the equation $y' = \frac{y^2}{1-xy}$ ($xy \neq 1$).

Substitute the obtained value for y' in the LHS of the above equation.

$$\Rightarrow y^2 + xyy' = y'$$

$$\Rightarrow y' - xyy' = y^2$$

$$\Rightarrow y'(1-xy) = y^2$$

$$\Rightarrow y' = \frac{y^2}{(1-xy)}$$

Therefore, the LHS of the equation is found to be equivalent to the right hand side of the equation.

As a result, the provided function is the differential equation's solution.

8. $y - \cos y = x : (y \sin y + \cos y + x)y' = 1.$

Ans: The given equation is $y - \cos y = x$.

Differentiating equation on both the sides with respect to x , the equation becomes,

$$\frac{d}{dx} y - \frac{d}{dx} \cos y = \frac{d}{dx} x$$

The derivative of $\cos x$ is $-\sin x$, whereas the derivative $\frac{d}{dx}(a) = 1$, where a is constant and the derivative of x is 1.

Thus,

$$\Rightarrow \frac{d}{dx} y - \frac{d}{dx} \cos y = \frac{d}{dx} x$$

$$\Rightarrow y' \sin y \cdot y' = 1$$

$$\Rightarrow y'(1 + \sin y) = 1$$

$$\Rightarrow y' = \frac{1}{1 + \sin y}$$

Now, consider the LHS of the equation $(y \sin y + \cos y + x)y' = 1$.

Substitute the obtained value for y' in the LHS of the above equation.

$$\Rightarrow (y \sin y + \cos y + x)y' = (y \sin y + \cos y + y - \cos y) \times \frac{1}{1 + \sin y}$$

$$\Rightarrow y(1 + \sin y) \cdot \frac{1}{\sin y} = y$$

Therefore, the LHS of the equation is found to be equivalent to the right hand side of the equation.

As a result, the provided function is the differential equation's solution.

9. $x + y = \tan^{-1} y : y^2 y' + y^2 + 1 = 0$.

Ans: The given equation is $x + y = \tan^{-1} y$.

Differentiating equation on both sides with respect to x , the equation becomes,

$$\frac{d}{dx}(x + y) = \frac{d}{dx}(\tan^{-1} y)$$

The derivative of $\tan^{-1} x$ is $\frac{1}{1+x^2}$ and the derivative of x is 1.

$$\Rightarrow \frac{d}{dx}(x + y) = \frac{d}{dx}(\tan^{-1} y)$$

$$\Rightarrow 1 + y' = \left[\frac{1}{1+y^2} \right] y'$$

$$\Rightarrow y' \left[\frac{1}{1+y^2} - 1 \right] = 1$$

$$\Rightarrow y' \left[\frac{1 - (1+y^2)}{1+y^2} \right] = 1$$

$$\Rightarrow y' \left[\frac{-y^2}{1+y^2} \right] = 1$$

$$\Rightarrow y' = \frac{(-1+y^2)}{y^2}$$

Now, consider the LHS of the equation $y^2 y' + y^2 + 1 = 0$.

Substitute the obtained value for y' in the LHS of the above equation.

$$\Rightarrow y^2 \left[\frac{-(1+y^2)}{y^2} \right] + y^2 + 1 = 0$$

$$\Rightarrow -1 - y^2 + y^2 + 1 = 0$$

Thus, $y^2 y' + y^2 + 1 = 0$

Therefore, the LHS of the equation is found to be equivalent to the right hand side of the equation.

As a result, the provided function is the differential equation's solution.

10. $y = \sqrt{a^2 - x^2}$ $x \in (-a, a) : x + y \frac{dy}{dx} = 0 (y \neq 0)$.

Ans: The given equation is $y = \sqrt{a^2 - x^2}$.

Differentiating equation on both the sides of with respect to x , the equation becomes,

$$y' = \frac{d}{dx} \left(\sqrt{a^2 - x^2} \right)$$

The derivative of $\sqrt{x}$ is $\frac{1}{2\sqrt{x}}$ and the derivative of x^n , $\frac{d}{dx}(x^n) = nx^{n-1}$, whereas the derivative

$\frac{d}{dx}(a) = 1$, where a is constant.

Thus,

$$\Rightarrow y' = \frac{d}{dx}(\sqrt{a^2 - x^2})$$

$$\Rightarrow y' = \frac{1}{2\sqrt{a^2 - x^2}} \cdot \frac{d}{dx}(a^2 - x^2)$$

$$\Rightarrow y' = \frac{-2x}{2\sqrt{a^2 - x^2}}$$

$$\Rightarrow y' = \frac{-x}{\sqrt{a^2 - x^2}}$$

Now, consider the left-hand side of the equation to be considered for evaluation, that is,
 $x + y \frac{dy}{dx} = 0$.

Substitute y for $\frac{-x}{\sqrt{a^2 - x^2}}$ and $\sqrt{a^2 - x^2}$ for y in the LHS of the above equation,

$$\Rightarrow x + y \frac{dy}{dx} = x + \sqrt{a^2 - x^2} \cdot \frac{-x}{\sqrt{a^2 - x^2}}$$

$$\Rightarrow y' = x - x$$

$$\Rightarrow y' = 0$$

Therefore, the LHS of the equation is found to be equivalent to the right hand side of the equation.

As a result, the provided function is the differential equation's solution.

11. The numbers of arbitrary constants in the general solution of a differential equation of fourth order are:

- A. 0
- B. 2
- C. 3
- D. 4

Ans: A differential equation solution is an expression for the dependent variable in terms of one or more independent variables that fulfils the relation. All potential solutions are included in the general solution, which usually includes arbitrary constants or arbitrary functions.

It is known that the number of constants in the general solution of a differential equation of order n is equal to the order of the differential equation.

As a result, the general equation of the fourth order differential equation has four constants. As a result, D is the right response.

12. The numbers of arbitrary constants in the particular solution of a differential equation of third order are:

- A. 3
- B. 2
- C. 1
- D. 0

Ans: The general solution becomes the particular solution of the problem when the arbitrary constant of the general solution obtains a unique value.

A differential equation's particular solution is derived by applying the boundary conditions.

Also, it is known that in a particular solution of a differential equation, there are no arbitrary constants found. Thus, the correct response for is (D) which is 0.

Exercise 9.3

1. Find the general solution for $\frac{dy}{dx} = \frac{1 - \cos x}{1 + \cos x}$.

Ans: The given differential equation is $\frac{dy}{dx} = \frac{1 - \cos x}{1 + \cos x}$.

Use trigonometric half-angle identities to simplify:

$$\frac{dy}{dx} = \frac{1 - \cos x}{1 + \cos x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2 \sin^2 \frac{x}{2}}{2 \cos^2 \frac{x}{2}}$$

$$\Rightarrow \frac{dy}{dx} = \tan^2 \frac{x}{2}$$

$$\Rightarrow \frac{dy}{dx} = \sec^2 \frac{x}{2} - 1$$

Separate the differentials and integrate:

$$\int dy = \int \left(\sec^2 \frac{x}{2} - 1 \right) dx$$

$$\Rightarrow y = \int \sec^2 \frac{x}{2} dx - \int dx$$

$$\Rightarrow y = 2 \tan \frac{x}{2} - x + C$$

Thus, the general solution of given differential equation is $y = 2 \tan \frac{x}{2} - x + C$.

2. Find the general solution for $\frac{dy}{dx} = \sqrt{4 - y^2} (-2 - y)$.

Ans: The given differential equation is $\frac{dy}{dx} = \sqrt{4 - y^2} (-2 - y)$.

Simplify the expression:

$$\frac{dy}{dx} = \sqrt{4 - y^2}$$

$$\Rightarrow \frac{dy}{\sqrt{4 - y^2}} = dx$$

Use standard integration:

$$\int \frac{dy}{\sqrt{4 - y^2}} = \int dx$$

$$\Rightarrow \sin^{-1} \frac{y}{2} = x + C$$

$$\Rightarrow \frac{y}{2} = \sin(x + C)$$

$$\Rightarrow y = 2 \sin(x + C)$$

Thus, the general solution of given differential equation is $y = 2 \sin(x + C)$.

3. Find the general solution for $\frac{dy}{dx} + y = 1 (y \neq 1)$.

Ans: The given differential equation is $\frac{dy}{dx} + y = 1 (y \neq 1)$.

Simplify the expression:

$$\frac{dy}{dx} + y = 1$$

$$\Rightarrow \frac{dy}{dx} = 1 - y$$

$$\Rightarrow \frac{dy}{1-y} = dx$$

Use standard integration:

$$\int \frac{dy}{1-y} = \int dx$$

$$\Rightarrow -\log(1-y) = x + C$$

$$\Rightarrow \log(1-y) = -(x + C)$$

$$\Rightarrow 1-y = e^{-(x+C)}$$

$$y = 1 - Ae^{-x} (A = e^{-C})$$

Thus, the general solution of given differential equation is $y = 1 - Ae^{-x}$.

4. Find the general solution for $\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0$.

Ans: The given differential equation is:

$$\sec^2 x \tan y dx + \sec^2 y \tan x dy = 0.$$

Divide both side by $\tan x \tan y$:

$$\frac{\sec^2 x \tan y dx + \sec^2 y \tan x dy}{\tan x \tan y} = 0$$

$$\Rightarrow \frac{\sec^2 x}{\tan x} dx + \frac{\sec^2 y}{\tan y} dy = 0$$

Integrate both side:

$$\int \frac{\sec^2 x}{\tan x} dx + \int \frac{\sec^2 y}{\tan y} dy = 0$$

$$\Rightarrow \int \frac{\sec^2 y}{\tan y} dy = - \int \frac{\sec^2 x}{\tan x} dx \quad \dots\dots(1)$$

Use a substitution method for integration. Substitute $\tan x = u$:

For integral on RHS:

$$\Rightarrow \tan x = u$$

$$\Rightarrow \sec^2 x dx = du$$

$$\Rightarrow \int \frac{\sec^2 x}{\tan x} dx = \int \frac{du}{u}$$

$$\Rightarrow \int \frac{\sec^2 x}{\tan x} dx = \log u$$

$$\Rightarrow \int \frac{\sec^2 x}{\tan x} dx = \log(\tan x)$$

Thus, evaluating result form (1):

$$\Rightarrow \log(\tan y) = -\log(\tan x) + \log(C)$$

$$\Rightarrow \log(\tan y) = \log\left(\frac{C}{\tan x}\right)$$

$$\Rightarrow \tan x \tan y = C$$

Thus, the general solution of given differential equation is $\tan x \tan y = C$.

5. Find the general solution for $(e^x + e^{-x})dy - (e^x - e^{-x})dx = 0$.

Ans: The given differential equation is:

$$(e^x + e^{-x})dy - (e^x - e^{-x})dx = 0.$$

Simplify the expression:

$$dy = \left[\frac{e^x - e^{-x}}{e^x + e^{-x}} \right] dx$$

Integrate both side:

$$\int dy = \int \left[\frac{e^x - e^{-x}}{e^x + e^{-x}} \right] dx \dots\dots(1)$$

Use a substitution method for integration. Substitute $e^x + e^{-x} = t$:

For integral on RHS:

$$\Rightarrow e^x + e^{-x} = t$$

$$\Rightarrow (e^x - e^{-x})dx = dt$$

$$\Rightarrow \int \left[\frac{e^x - e^{-x}}{e^x + e^{-x}} \right] dx = \int \frac{dt}{t} \quad s$$

$$\Rightarrow \int \left[\frac{e^x - e^{-x}}{e^x + e^{-x}} \right] dx = \ln t + C$$

$$\Rightarrow \int \left[\frac{e^x - e^{-x}}{e^x + e^{-x}} \right] dx = \log(e^x + e^{-x}) + C$$

Thus, evaluating result form (1):

$$y = \log(e^x + e^{-x}) + C$$

Thus, the general solution of given differential equation is $y = \log(e^x + e^{-x}) + C$.

6. Find the general solution for $\frac{dy}{dx} = (1+x^2)(1+y^2)$.

Ans: The given differential equation is:

$$\frac{dy}{dx} = (1+x^2)(1+y^2).$$

Simplify the expression:

$$\frac{dy}{1+y^2} = (1+x^2)dx$$

Integrate both side:

$$\int \frac{dy}{1+y^2} = \int (1+x^2)dx$$

Use standard integration:

$$\tan^{-1} y = \int dx + \int x^2 dx$$

$$\Rightarrow \tan^{-1} y = x + \frac{x^3}{3} + C$$

Thus, the general solution of given differential equation is $\tan^{-1} y = x + \frac{x^3}{3} + C$.

7. Find the general solution for $y \log y \, dx - x \, dy = 0$.

Ans: The given differential equation is:

$$y \log y \, dx - x \, dy = 0.$$

Simplify the expression:

$$y \log y \, dx = x \, dy$$

$$\Rightarrow \frac{dx}{x} = \frac{dy}{y \log y}$$

Integrate both side:

$$\int \frac{dy}{y \log y} = \int \frac{dx}{x} \quad \dots\dots(1)$$

Use substitution method for integration on LHS. Substitute $\log y = t$:

$$\log y = t$$

$$\Rightarrow \frac{1}{y} dy = dt$$

$$\Rightarrow \int \frac{dy}{y \log y} = \int \frac{dt}{t}$$

$$\Rightarrow \int \frac{dy}{y \log y} = \log t$$

$$\Rightarrow \int \frac{dy}{y \log y} = \log(\log y)$$

Evaluating expression (1):

$$\log(\log y) = \log x + \log C$$

$$\Rightarrow \log(\log y) = \log Cx$$

$$\Rightarrow \log y = Cx$$

$$\Rightarrow y = e^{Cx}$$

Thus, the general solution of given differential equation is $y = e^{Cx}$.

8. Find the general solution for $x^5 \frac{dy}{dx} = -y^5$.

Ans: The given differential equation is:

$$x^5 \frac{dy}{dx} = -y^5.$$

Simplify the expression:

$$\frac{dy}{y^5} = -\frac{dx}{x^5}$$

Integrate both side:

$$\int \frac{dy}{y^5} = -\int \frac{dx}{x^5}$$

$$\Rightarrow \int y^{-5} dy = -\int x^{-5} dx$$

$$\Rightarrow \frac{y^{-5+1}}{-5+1} = -\frac{x^{-5+1}}{-5+1} + C$$

$$\Rightarrow \frac{y^{-4}}{-4} = -\frac{x^{-4}}{-4} + C$$

$$\Rightarrow x^{-4} + y^{-4} = -4C$$

$$\Rightarrow x^{-4} + y^{-4} = A \quad (A = -4C)$$

Thus, the general solution of given differential equation is $x^{-4} + y^{-4} = A$.

9. Find the general solution for $\frac{dy}{dx} = \sin^{-1} x$.

Ans: The given differential equation is:

$$\frac{dy}{dx} = \sin^{-1} x.$$

Simplify the expression:

$$dy = \sin^{-1} x dx$$

Integrate both side:

$$\int dy = \int \sin^{-1} x dx$$

$$\Rightarrow y = \int 1 \times \sin^{-1} x dx$$

Use product rule of integration:

$$\int \sin^{-1} x dx = \sin^{-1} x \int dx - \int \left(\frac{1}{\sqrt{1-x^2}} \int dx \right) dx$$

$$\Rightarrow \int \sin^{-1} x dx = x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx$$

Substitute $1-x^2=t^2$

$$1-x^2=t^2$$

$$\Rightarrow -2x dx = 2t dt$$

$$\Rightarrow -x dx = t dt$$

Evaluating the integral:

$$\frac{2x^2 + x}{(x+1)(x^2+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2+1}$$

$$\Rightarrow \int \sin^{-1} x dx = x \sin^{-1} x + \int \frac{t dt}{\sqrt{t^2}}$$

$$\Rightarrow \int \sin^{-1} x dx = x \sin^{-1} x + t + C$$

$$\Rightarrow \int \sin^{-1} x dx = x \sin^{-1} x + \sqrt{1-x^2} + C$$

$$\Rightarrow y = x \sin^{-1} x + \sqrt{1-x^2} + C$$

Thus, the general solution of given differential equation is $y = x \sin^{-1} x + \sqrt{1-x^2} + C$.

10. Find the general solution for $e^x \tan y dx + (1-e^x) \sec^2 y dy = 0$.

Ans: The given differential equation is:

$$e^x \tan y dx + (1 - e^x) \sec^2 y dy = 0.$$

Simplify the expression:

$$(1 - e^x) \sec^2 y dy = -e^x \tan y dx$$

$$\Rightarrow \frac{\sec^2 y}{\tan y} dy = -\frac{e^x}{(1 - e^x)} dx$$

Integrate both side:

$$\int \frac{\sec^2 y}{\tan y} dy = -\int \frac{e^x}{(1 - e^x)} dx \quad \dots\dots(1)$$

Substitute $\tan y = u$

$$\tan y = u$$

$$\Rightarrow \sec^2 y = du$$

Evaluating the LHS integral of (1):

$$\Rightarrow \int \frac{\sec^2 y}{\tan y} dy = \int \frac{du}{u}$$

$$\Rightarrow \int \frac{\sec^2 y}{\tan y} dy = \log u$$

$$\Rightarrow \int \frac{\sec^2 y}{\tan y} dy = \log(\tan y)$$

Substitute $1 - e^x = v$

$$1 - e^x = v$$

$$\Rightarrow -e^x dx = dv$$

Evaluating the RHS integral of (1):

$$\Rightarrow -\int \frac{e^x}{(1-e^x)} dx = \int \frac{dv}{v}$$

$$\Rightarrow -\int \frac{e^x}{(1-e^x)} dx = \log v$$

$$\Rightarrow -\int \frac{e^x}{(1-e^x)} dx = \log(1-e^x)$$

Therefore, the integral (1) will be:

$$\log(\tan y) = \log(1-e^x) + \log C$$

$$\Rightarrow \log(\tan y) = \log C(1-e^x)$$

$$\Rightarrow \tan y = C(1-e^x)$$

Thus, the general solution of given differential equation is $\tan y = C(1-e^x)$.

11. Find the particular solution of $(x^3 + x^2 + x + 1) \frac{dy}{dx} = 2x^2 + x$; $y=1, x=0$ to satisfy the given condition.

Ans: The given differential equation is:

$$(x^3 + x^2 + x + 1) \frac{dy}{dx} = 2x^2 + x; y=1, x=0.$$

Simplify the expression:

$$\frac{dy}{dx} = \frac{2x^2 + x}{(x^3 + x^2 + x + 1)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x^2 + x}{(x^3 + x + x^2 + 1)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{2x^2 + x}{(x+1)(x^2 + 1)}$$

$$\Rightarrow dy = \frac{2x^2 + x}{(x+1)(x^2 + 1)} dx$$

Integrate both side:

$$\int dy = \int \frac{2x^2 + x}{(x+1)(x^2 + 1)} dx \quad \dots\dots(1)$$

Use partial fraction method to simplify the RHS:

$$\frac{2x^2 + x}{(x+1)(x^2 + 1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2 + 1}$$

$$\Rightarrow 2x^2 + x = A(x^2 + 1) + (Bx + C)(x + 1)$$

$$\Rightarrow 2x^2 + x = (A + B)x^2 + (B + C)x + (A + C)$$

By comparing coefficients:

$$A + B = 2$$

$$B + C = 1$$

$$A + C = 0$$

Solving this we get:

$$\frac{2x^2 + x}{(x+1)(x^2+1)} = \frac{\left(\frac{1}{2}\right)}{x+1} + \frac{\left(\frac{3}{2}\right)x + \left(-\frac{1}{2}\right)}{x^2+1}$$

$$\Rightarrow \frac{2x^2 + x}{(x+1)(x^2+1)} = \frac{1}{2} \left(\frac{1}{x+1} + \frac{3x-1}{x^2+1} \right)$$

Rewriting the integral(1):

$$y = \frac{1}{2} \int \left(\frac{1}{x+1} + \frac{3x-1}{x^2+1} \right) dx$$

$$\Rightarrow y = \frac{1}{2} \int \frac{1}{x+1} dx + \frac{1}{2} \int \frac{3x-1}{x^2+1} dx$$

$$\Rightarrow y = \frac{1}{2} \log(x+1) + \frac{3}{2} \int \frac{x}{x^2+1} dx - \frac{1}{2} \int \frac{1}{x^2+1} dx$$

$$\Rightarrow y = \frac{1}{2} \log(x+1) + \frac{3}{4} \int \frac{2x}{x^2+1} dx - \frac{1}{2} \tan^{-1} x$$

$$\Rightarrow y = \frac{1}{2} \log(x+1) + \frac{3}{4} \log(x^2+1) - \frac{1}{2} \tan^{-1} x + C$$

For $y=1$ when $x=0$.

$$1 = \frac{1}{2} \log(0+1) + \frac{3}{4} \log(0+1) - \frac{1}{2} \tan^{-1} 0 + C$$

$$\Rightarrow C = 1$$

Thus, the required particular solution is: $\Rightarrow y = \frac{1}{2} \log(x+1) + \frac{3}{4} \log(x^2+1) - \frac{1}{2} \tan^{-1} x + 1$.

12. Find the particular solution of $x(x^2-1)\frac{dy}{dx}=1$; $y=0$ when $x=2$ to satisfy the given condition.

Ans: The given differential equation is:

$$x(x^2-1)\frac{dy}{dx}=1; y=0 \text{ when } x=2$$

Simplify the expression:

$$x(x^2-1)\frac{dy}{dx}=1$$

$$\Rightarrow dy = \frac{dx}{x(x^2-1)}$$

$$\Rightarrow dy = \frac{dx}{x(x-1)(x+1)}$$

Integrate both side:

$$\int dy = \int \frac{dx}{x(x-1)(x+1)} \quad \dots\dots(1)$$

Use partial fraction method to simplify the RHS:

$$\frac{1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$$

$$\Rightarrow 1 = A(x^2-1) + Bx(x+1) + Cx(x-1)$$

$$\Rightarrow 1 = (A+B+C)x^2 + (B-C)x - A$$

By comparing coefficients:

$$A+B+C=0$$

$$B - C = 0$$

$$-A = 1$$

Solving this we get:

$$\frac{1}{x(x-1)(x+1)} = \frac{(-1)}{x} + \frac{\left(\frac{1}{2}\right)}{x-1} + \frac{\left(\frac{1}{2}\right)}{x+1}$$

$$\Rightarrow \frac{1}{x(x-1)(x+1)} = -\frac{1}{x} + \frac{1}{2} \left(\frac{1}{x-1} + \frac{1}{x+1} \right)$$

Rewriting the integral(1):

$$y = \int \left(-\frac{1}{x} + \frac{1}{2} \left(\frac{1}{x-1} + \frac{1}{x+1} \right) \right) dx$$

$$\Rightarrow y = -\int \frac{1}{x} dx + \frac{1}{2} \int \frac{1}{x-1} dx + \frac{1}{2} \int \frac{1}{x+1} dx$$

$$\Rightarrow y = -\log x + \frac{1}{2} \log(x-1) + \frac{1}{2} \log(x+1) + \log C$$

$$\Rightarrow y = -\frac{2}{2} \log x + \frac{1}{2} \log(x-1) + \frac{1}{2} \log(x+1) + \frac{2}{2} \log C$$

$$\Rightarrow y = \frac{1}{2} \left(-\log x^2 + \log(x-1) + \log(x+1) + \log C^2 \right)$$

$$\Rightarrow y = \frac{1}{2} \log \left[\frac{C^2(x^2-1)}{x^2} \right]$$

For $y = 0$ when $x=2$.

$$0 = \frac{1}{2} \log \left[\frac{C^2(2^2 - 1)}{2^2} \right]$$

$$\Rightarrow 0 = \log \left[\frac{3C^2}{4} \right]$$

$$\Rightarrow \frac{3C^2}{4} = 1$$

$$\Rightarrow C^2 = \frac{4}{3}$$

Thus, the required particular solution is: $y = \frac{1}{2} \log \left[\frac{4(x^2 - 1)}{3x^2} \right]$.

13. Find the particular solution of $\cos \left(\frac{dy}{dx} \right) = a$ ($a \in \mathbb{R}$); $y=1$ when $x=0$ to satisfy the given condition.

Ans: The given differential equation is:

$$\cos \left(\frac{dy}{dx} \right) = a \quad (a \in \mathbb{R}); \quad y=1 \text{ when } x=0$$

Simplify the expression:

$$\cos \left(\frac{dy}{dx} \right) = a$$

$$\Rightarrow \frac{dy}{dx} = \cos^{-1} a$$

$$\Rightarrow dy = \cos^{-1} a \, dx$$

Integrate both side:

$$\int dy = \int \cos^{-1} a dx$$

$$\Rightarrow y = \cos^{-1} a \int dx$$

$$\Rightarrow y = x \cos^{-1} a + C$$

For $y=1$ when $x=0$.

$$1 = 0 \cos^{-1} a + C$$

$$\Rightarrow C = 1$$

Thus, the required particular solution is:

$$y = x \cos^{-1} a + 1$$

$$\Rightarrow \frac{y-1}{x} = \cos^{-1} a$$

$$\Rightarrow \cos\left(\frac{y-1}{x}\right) = a$$

14. Find the particular solution of $\frac{dy}{dx} = y \tan x$; $y=1$ when $x=0$ to satisfy the given condition.

Ans: The given differential equation is:

$$\frac{dy}{dx} = y \tan x; y=1 \text{ when } x=0$$

Simplify the expression:

$$\frac{dy}{dx} = y \tan x$$

$$\Rightarrow \frac{dy}{y} = \tan x dx$$

Integrate both side:

$$\int \frac{dy}{y} = \int \tan x dx$$

$$\Rightarrow \log y = \log(\sec x) + \log C$$

$$\Rightarrow \log y = \log(C \sec x)$$

$$\Rightarrow y = C \sec x$$

For $y=1$ when $x=0$.

$$1 = C \sec 0$$

$$\Rightarrow C = 1$$

Thus, the required particular solution is: $y = \sec x$.

15. Find the equation of a curve passing through the point (0, 0) and whose differential equation is (Tex translation failed).

Ans: The given differential equation is: $y' = e^x \sin x$

The curve passes through (0,0).

Simplify the expression:

$$\Rightarrow \frac{dy}{dx} = e^x \sin x$$

$$\Rightarrow dy = e^x \sin x dx$$

Integrate both side:

$$\int dy = \int e^x \sin x dx$$

Use product rules for integration of RHS. Let:

$$I = \int e^x \sin x dx$$

$$\Rightarrow I = \sin x \int e^x dx - \int (\cos x \int e^x dx) dx$$

$$\Rightarrow I = e^x \sin x - \int e^x \cos x dx$$

$$\Rightarrow I = e^x \sin x - \left(\cos x \int e^x dx + \int (\sin x \int e^x dx) dx \right)$$

$$\Rightarrow I = e^x \sin x - e^x \cos x - \int (e^x \sin x) dx$$

$$\Rightarrow I = e^x \sin x - e^x \cos x - I$$

$$\Rightarrow I = \frac{e^x}{2} (\sin x - \cos x)$$

Thus integral will be:

$$y = \frac{e^x}{2} (\sin x - \cos x) + C$$

Thus, as the curve passes through (0,0)

$$0 = \frac{e^0}{2} (\sin 0 - \cos 0) + C$$

$$\Rightarrow 0 = \frac{1}{2}(0-1) + C$$

$$\Rightarrow C = \frac{1}{2}$$

Thus, the equation of the curve will be:

$$y = \frac{e^x}{2}(\sin x - \cos x) + \frac{1}{2}$$

$$\Rightarrow y = \frac{e^x}{2}(\sin x - \cos x + 1)$$

16. For the differential equation $xy \frac{dy}{dx} = (x+2)(y+2)$ find the solution curve passing through the point $(1, -1)$.

Ans: The given differential equation is:

$$xy \frac{dy}{dx} = (x+2)(y+2)$$

The curve passes through $(1, -1)$.

Simplify the expression:

$$\Rightarrow \left(\frac{y}{y+2} \right) dy = \frac{(x+2)}{x} dx$$

$$\Rightarrow \left(1 - \frac{2}{y+2} \right) dy = \frac{(x+2)}{x} dx$$

Integrate both side:

$$\int \left(1 - \frac{2}{y+2}\right) dy = \int \frac{(x+2)}{x} dx$$

$$\Rightarrow \int dy - 2 \int \frac{1}{y+2} dy = \int \frac{x}{x} dx + \int \frac{2}{x} dx$$

$$\Rightarrow y - 2 \log(y+2) = x + 2 \log x + C$$

$$\Rightarrow y - x = 2 \log(y+2) + 2 \log x + C$$

$$\Rightarrow y - x = 2 \log[x(y+2)] + C$$

$$\Rightarrow y - x = \log[x^2(y+2)^2] + C$$

Thus, as the curve passes through (1,-1)

$$\Rightarrow -1 - 1 = \log[(1)^2(-1+2)^2] + C$$

$$\Rightarrow -2 = \log 1 + C$$

$$\Rightarrow C = -2$$

Thus, the equation of the curve will be:

$$y - x = \log[x^2(y+2)^2] - 2$$

$$\Rightarrow y - x + 2 = \log(x^2(y+2)^2)$$

17. Find the equation of a curve passing through the point (0,-2) given that at any point (x,y) on the curve, the product of the slope of its tangent and y-coordinate of the point is equal to the x-coordinate of the point.

Ans: According to , the equation is given by: $y \frac{dy}{dx} = x$

The curve passes through $(0, -2)$.

Simplify the expression:

$$\Rightarrow ydy = xdx$$

Integrate both side:

$$\int ydy = \int xdx$$

$$\Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C$$

$$\Rightarrow y^2 - x^2 = 2C$$

Thus, as the curve passes through $(0, -2)$

$$\Rightarrow (-2)^2 - 0^2 = 2C$$

$$\Rightarrow 4 = 2C$$

$$\Rightarrow C = 2$$

Thus, the equation of the curve will be:

$$y^2 - x^2 = 2(2)$$

$$\Rightarrow y^2 - x^2 = 4 .$$

18. At any point (x,y) of a curve, the slope of the tangent is twice the slope of the line segment joining the point of contact to the point $(-4,-3)$. Find the equation of the curve given that it passes through $(-2,1)$.

Ans: Let the point of contact of the tangent be (x,y) . Then the slope of the segment joining point of contact and $(-4,-3)$:

$$m = \frac{y+3}{x+4}$$

According to the for the slope of tangent $\frac{dy}{dx}$ it follows:

$$\frac{dy}{dx} = 2m$$

$$\Rightarrow \frac{dy}{dx} = 2\left(\frac{y+3}{x+4}\right)$$

Simplify the expression:

$$\frac{dy}{dx} = 2\left(\frac{y+3}{x+4}\right)$$

$$\frac{dy}{y+3} = \frac{2}{x+4} dx$$

Integrate both side:

$$\int \frac{dy}{y+3} = \int \frac{2}{x+4} dx$$

$$\Rightarrow \log(y+3) = 2\log(x+4) + \log C$$

$$\Rightarrow \log(y+3) = \log(x+4)^2 + \log C$$

$$\Rightarrow \log(y+3) = \log C(x+4)^2$$

$$\Rightarrow y+3 = C(x+4)^2$$

Thus, as the curve passes through $(-2,1)$

$$1+3 = C(-2+4)^2$$

$$\Rightarrow 4 = 4C$$

$$\Rightarrow C = 1$$

Thus, the equation of the curve will be:

$$y+3 = (x+4)^2$$

19. The volume of spherical balloons being inflated changes at a constant rate. If initially its radius is 3 units and after 3 seconds it is 6 units. Find the radius of balloon after t seconds.

Ans: Let the volume of spherical balloon be V and its radius r . Let the rate of change of volume be k .

$$\frac{dV}{dt} = k$$

$$\Rightarrow \frac{d}{dt} \left(\frac{4}{3} \pi r^3 \right) = k$$

$$\Rightarrow \frac{4}{3} \pi \frac{d}{dt} (r^3) = k$$

$$\Rightarrow \frac{4}{3} \pi (3r^2) \frac{dr}{dt} = k$$

$$\Rightarrow 4\pi r^2 \frac{dr}{dt} = k$$

$$\Rightarrow 4\pi r^2 dr = k dt$$

Integrate both side:

$$\int 4\pi r^2 dr = \int k dt$$

$$\Rightarrow 4\pi \int r^2 dr = kt + C$$

$$\Rightarrow \frac{4}{3} \pi r^3 = kt + C$$

At initial time, $t=0$ and $r=3$:

$$\frac{4}{3} \pi 3^3 = k(0) + C$$

$$\Rightarrow C = 36\pi$$

At $t=3$ the radius $r=6$:

$$\frac{4}{3} \pi (6^3) = k(3) + 36\pi$$

$$\Rightarrow 3k = 288\pi - 36\pi$$

$$\Rightarrow k = 84\pi$$

Thus, the radius-time relation can be given by:

$$\frac{4}{3} \pi r^3 = 84\pi t + 36\pi$$

$$\Rightarrow r^3 = 63t + 27$$

$$\Rightarrow r = (63t + 27)^{\frac{1}{3}}$$

The radius of balloon after t seconds is given by: $r = (63t + 27)^{\frac{1}{3}}$

20. In a bank, principal increases continuously at the rate of r % per year. Find the value of r if Rs. 100 doubles itself in 10 years ($\log_e 2 = 0.6931$).

Ans: Let the principal be p , according to :

$$\frac{dp}{dt} = \left(\frac{r}{100} \right) p$$

Simplify the expression:

$$\frac{dp}{p} = \left(\frac{r}{100} \right) dt$$

Integrate both side:

$$\int \frac{dp}{p} = \int \left(\frac{r}{100} \right) dt$$

$$\Rightarrow \log p = \frac{rt}{100} + c$$

$$\Rightarrow p = e^{\frac{rt}{100} + c}$$

$$\Rightarrow p = Ae^{\frac{rt}{100}} \quad (A = e^c)$$

At $t=0$, $p=100$:

$$100 = Ae^{\frac{r(0)}{100}}$$

$$\Rightarrow A = 100$$

Thus, the principle and rate of interest relation:

$$p = 100e^{\frac{rt}{100}}$$

At $t=10$, $p=2 \times 100=200$:

$$200 = 100e^{\frac{r(10)}{100}}$$

$$\Rightarrow 2 = e^{\frac{r}{10}}$$

Take logarithm on both side:

$$\log\left(e^{\frac{r}{10}}\right) = \log(2)$$

$$\Rightarrow \frac{r}{10} = 0.6931$$

$$\Rightarrow r = 6.931$$

Thus, the rate of interest $r=6.931\%$.

21. In a bank, principal increases continuously at the rate of 5 % per year. An amount of Rs 1000 is deposited with this bank, how much will it worth after 10 years ($e^{0.5}=1.648$).

Ans: Let the principal be p , according to principle increases at the rate of 5 % per year.

$$\frac{dp}{dt} = \left(\frac{5}{100}\right)p$$

Simplify the expression:

$$\frac{dp}{dt} = \frac{p}{20}$$

$$\Rightarrow \frac{dp}{p} = \frac{1}{20} dt$$

Integrate both side:

$$\int \frac{dp}{p} = \int \frac{1}{20} dt$$

$$\Rightarrow \log p = \frac{t}{20} + C$$

$$\Rightarrow p = e^{\frac{t}{20} + C}$$

$$\Rightarrow p = Ae^{\frac{t}{20}} \left(A = e^C \right)$$

At $t=0$, $p=1000$:

$$1000 = Ae^{\frac{0}{20}}$$

$$\Rightarrow A = 1000$$

Thus, the relation of principal and time relation:

$$\Rightarrow p = 1000e^{\frac{t}{20}}$$

At $t=10$:

$$p = 1000e^{\frac{10}{20}}$$

$$\Rightarrow p = 1000e^{0.5}$$

$$\Rightarrow p = 1000 \times 1.648$$

$$\Rightarrow p = 1648$$

Thus after 10 this year the amount will become Rs. 1648.

22. In a culture, the bacteria count is 100000 . The number is increased by 10 % in 2 hours. In how many hours will the count reach 200000 , if the rate of growth of bacteria is proportional to the number present?

Ans: Let the number of bacteria be y at time t . According to:

$$\frac{dy}{dt} \propto y$$

$$\frac{dy}{dt} = cy$$

Here c is constant.

Simplify the expression:

$$\frac{dy}{y} = cdt$$

Integrate both side:

$$\int \frac{dy}{y} = \int cdt$$

$$\Rightarrow \log y = ct + D$$

$$\Rightarrow y = e^{ct+D}$$

$$\Rightarrow y = Ae^{ct} \quad (A = e^D)$$

At $t=0$, $y=100000$:

$$100000 = Ae^{c(0)}$$

$$\Rightarrow A = 100000$$

$$\text{At } t=2, y = \frac{11}{10}(100000) = 110000:$$

$$y = 100000e^{ct}$$

$$\Rightarrow 110000 = 100000e^{c(2)}$$

$$\Rightarrow e^{2c} = \frac{11}{10}$$

$$\Rightarrow 2c = \log\left(\frac{11}{10}\right)$$

$$\Rightarrow c = \frac{1}{2}\log\left(\frac{11}{10}\right) \quad \dots\dots(1)$$

For $y=200000$:

$$200000 = 100000e^{ct}$$

$$\Rightarrow e^{ct} = 2$$

$$\Rightarrow ct = \log 2$$

$$\Rightarrow t = \frac{\log 2}{c}$$

Back substituting using expression (1):

$$t = \frac{\log 2}{\frac{1}{2}\log\left(\frac{11}{10}\right)}$$

$$\Rightarrow t = \frac{2\log 2}{\log\left(\frac{11}{10}\right)}$$

Thus, time required for bacteria to reach 200000 is

$$t = \frac{\log 2}{\frac{1}{2} \log \left(\frac{11}{10} \right)} \text{ hrs.}$$

23. Find the general solution of the differential equation $\frac{dy}{dx} = e^{x+y}$.

(A) $e^x + e^{-y} = C$

(B) $e^x + e^y = C$

(C) $e^{-x} + e^y = C$

(D) $e^{-x} + e^{-y} = C$

Ans: The given differential equation is $\frac{dy}{dx} = e^{x+y}$. Simplify the expression:

$$\frac{dy}{dx} = e^x e^y$$

$$\Rightarrow \frac{dy}{e^y} = e^x dx$$

$$\Rightarrow e^{-y} dy = e^x dx$$

Integrate both side:

$$\int e^{-y} dy = \int e^x dx$$

$$\Rightarrow -e^{-y} = e^x + D$$

$$\Rightarrow e^x + e^{-y} = -D$$

$$\Rightarrow e^x + e^{-y} = C \quad (C = -D)$$

Thus, the general solution of given differential equation is $e^x + e^{-y} = C$

Thus, the correct option is (A).

Exercise 9.4

1. Show that, differential equation $(x^2 + xy)dy = (x^2 + y^2)dx$ is homogenous and solves it.

Ans: Rewrite the equation in standard form:

$$\frac{dy}{dx} = \frac{x^2 + y^2}{x^2 + xy}$$

Checking for homogeneity:

$$F(x, y) = \frac{x^2 + y^2}{x^2 + xy}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{(\lambda x)^2 + (\lambda y)^2}{(\lambda x)^2 + (\lambda x)(\lambda y)}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda^2(x^2 + y^2)}{\lambda^2(x^2 + xy)}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{x^2 + y^2}{x^2 + xy}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y=vx$:

$$\frac{d(vx)}{dx} = \frac{x^2 + (vx)^2}{x^2 + x(vx)}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = \frac{x^2(1+v^2)}{x^2(1+v)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1+v^2}{1+v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+v^2}{1+v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+v^2 - v - v^2}{1+v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1-v}{1+v}$$

Separate the differentials:

$$\frac{1+v}{1-v} dv = \frac{dx}{x}$$

Integrate both side:

$$\int \frac{1+v}{1-v} dv = \int \frac{dx}{x}$$

$$\Rightarrow \int \frac{2-(1-v)}{1-v} dv = \log x - \log k$$

$$\Rightarrow \int \frac{2}{1-v} dv - \int \frac{1-v}{1-v} dv = \log \frac{x}{k}$$

$$\Rightarrow -2\log(1-v) - \int dv = \log \frac{x}{k}$$

$$\Rightarrow -2\log(1-v) - v = \log \frac{x}{k}$$

$$\Rightarrow v = -\log \frac{x}{k} - 2\log(1-v)$$

$$\Rightarrow v = \log \left(\frac{k}{x(1-v)^2} \right)$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow \frac{y}{x} = \log \left(\frac{k}{x \left(1 - \frac{y}{x} \right)^2} \right)$$

$$\Rightarrow \frac{y}{x} = \log \left(\frac{kx}{(x-y)^2} \right)$$

$$\Rightarrow e^{\frac{y}{x}} = \frac{kx}{(x-y)^2}$$

$$\Rightarrow (x-y)^2 = kxe^{-\frac{y}{x}}$$

The solution of the given differential equation $(x-y)^2 = kxe^{-\frac{y}{x}}$.

2. Show that, differential equation $y' = \frac{x+y}{x}$ is homogenous and solves it.

Ans: Rewrite the equation in standard form:

$$\frac{dy}{dx} = \frac{x+y}{x}$$

Checking for homogeneity:

$$F(x, y) = \frac{x+y}{x}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda x + \lambda y}{\lambda x}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda(x+y)}{\lambda(x)}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{x+y}{x}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y=vx$:

$$\frac{d(vx)}{dx} = \frac{x+(vx)}{x}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = \frac{x(1+v)}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = 1+v$$

$$\Rightarrow x \frac{dv}{dx} = 1$$

Separate the differentials:

$$dv = \frac{dx}{x}$$

Integrate both side:

$$\int dv = \int \frac{dx}{x}$$

$$\Rightarrow \int dv = \log x + \log k$$

$$\Rightarrow v = \log kx$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow \frac{y}{x} = \log kx$$

$$\Rightarrow y = x \log kx$$

The solution of the given differential equation $y = x \log kx$

3. Show that, differential equation $(x-y)dy - (x+y)dx = 0$ is homogenous and solves it.

Ans: Rewrite the equation in standard form:

$$\frac{dy}{dx} = \frac{x+y}{x-y}$$

Checking for homogeneity:

$$F(x, y) = \frac{x+y}{x-y}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda x + \lambda y}{\lambda x - \lambda y}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda(x+y)}{\lambda(x-y)}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{x+y}{x-y}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y=vx$:

$$\frac{d(vx)}{dx} = \frac{x+(vx)}{x-vx}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = \frac{x(1+v)}{x(1-v)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1+v}{1-v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+v}{1-v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+v-v+v^2}{1-v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+v^2}{1-v}$$

Separate the differentials:

$$\frac{1-v}{1+v^2} dv = \frac{dx}{x}$$

Integrate both side:

$$\int \frac{1-v}{1+v^2} dv = \int \frac{dx}{x}$$

$$\Rightarrow \int \frac{1-v}{1+v^2} dv = \int \frac{dx}{x}$$

$$\Rightarrow \int \frac{1}{1+v^2} dv - \int \frac{v}{1+v^2} dv = \log x + C$$

$$\Rightarrow \tan^{-1} v - \frac{1}{2} \int \frac{2v}{1+v^2} dv = \log x + C$$

$$\Rightarrow \tan^{-1} v - \frac{1}{2} \log(1+v^2) = \log x + C$$

$$\Rightarrow \tan^{-1} v = \log x + \frac{1}{2} \log(1+v^2) + C$$

$$\Rightarrow \tan^{-1} v = \frac{1}{2} \log x^2 + \frac{1}{2} \log(1+v^2) + C$$

$$\Rightarrow \tan^{-1} v = \frac{1}{2} \log [x^2 (1+v^2)] + C$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow \tan^{-1} \frac{y}{x} = \frac{1}{2} \log \left[x^2 \left(1 + \frac{y^2}{x^2} \right) \right] + C$$

$$\Rightarrow \tan^{-1} \frac{y}{x} = \frac{1}{2} \log (x^2 + y^2) + C$$

The solution of the given differential equation $\tan^{-1} \frac{y}{x} = \frac{1}{2} \log (x^2 + y^2) + C$.

4. Show that, differential equation $(x^2 - y^2)dx + 2xy dy = 0$ is homogenous and solves it.

Ans: Rewrite the equation in standard form:

$$\frac{dy}{dx} = -\frac{x^2 - y^2}{2xy}$$

Checking for homogeneity:

$$F(x, y) = -\frac{x^2 - y^2}{2xy}$$

$$\Rightarrow F(\lambda x, \lambda y) = -\frac{(\lambda x)^2 - (\lambda y)^2}{2(\lambda x)(\lambda y)}$$

$$\Rightarrow F(\lambda x, \lambda y) = -\frac{\lambda^2 x^2 - \lambda^2 y^2}{2\lambda^2 xy}$$

$$\Rightarrow F(\lambda x, \lambda y) = -\frac{\lambda^2 (x^2 - y^2)}{\lambda^2 (2xy)}$$

$$\Rightarrow F(\lambda x, \lambda y) = -\frac{x^2 - y^2}{2xy}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y = vx$:

$$\frac{d(vx)}{dx} = -\frac{x^2 - (vx)^2}{2x(vx)}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = \frac{x^2(v^2 - 1)}{x^2(2v)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v^2 - 1}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 - 1 - 2v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = -\frac{1 + v^2}{2v}$$

Separate the differentials:

$$\frac{2v}{1 + v^2} dv = -\frac{dx}{x}$$

Integrate both side:

$$\int \frac{2v}{1 + v^2} dv = -\int \frac{dx}{x}$$

$$\Rightarrow \log(1 + v^2) = -\log x + C$$

$$\Rightarrow \log(1 + v^2) + \log x = C$$

$$\Rightarrow \log[x(1 + v^2)] = C$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow \log \left[x \left(1 + \frac{y^2}{x^2} \right) \right] = C$$

$$\Rightarrow \left(\frac{x^2 + y^2}{x} \right) = k \quad k = e^C$$

$$\Rightarrow x^2 + y^2 = kx$$

The solution of the given differential equation $x^2 + y^2 = kx$.

5. Show that, differential equation $x^2 \frac{dy}{dx} = x^2 - 2y^2 + xy$ is homogenous and solves it.

Ans: Rewrite the equation in standard form: $\frac{dy}{dx} = \frac{x^2 - 2y^2 + xy}{x^2}$

Checking for homogeneity:

$$F(x, y) = \frac{x^2 - 2y^2 + xy}{x^2}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda^2 x^2 - 2\lambda^2 y^2 + (\lambda x)(\lambda y)}{\lambda^2 x^2}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda^2 (x^2 - 2y^2 + xy)}{\lambda^2 (x^2)}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{x^2 - 2y^2 + xy}{x^2}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y=vx$:

$$\frac{d(vx)}{dx} = \frac{x^2 - 2(vx)^2 + x(vx)}{x^2}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = \frac{x^2(1-2v^2+v)}{x^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = 1 - 2v^2 + v$$

$$\Rightarrow x \frac{dv}{dx} = 1 - 2v^2$$

Separate the differentials:

$$\frac{1}{1-2v^2} dv = \frac{dx}{x}$$

Integrate both side:

$$\int \frac{1}{1-2v^2} dv = \int \frac{dx}{x}$$

$$\Rightarrow \frac{1}{2} \int \frac{1}{\frac{1}{2} - v^2} dv = \log x + C$$

$$\Rightarrow \frac{1}{2} \int \frac{1}{\left(\frac{1}{\sqrt{2}}\right)^2 - v^2} dv = \log x + C$$

$$\Rightarrow \frac{1}{2} \left(\frac{1}{2 \times \frac{1}{\sqrt{2}}} \right) \log \left| \frac{\frac{1}{\sqrt{2}} + v}{\frac{1}{\sqrt{2}} - v} \right| = \log x + C$$

$$\Rightarrow \frac{1}{2\sqrt{2}} \log \left| \frac{1+\sqrt{2}v}{1-\sqrt{2}v} \right| = \log x + C$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow \frac{1}{2\sqrt{2}} \log \left| \frac{1+\sqrt{2}\left(\frac{y}{x}\right)}{1-\sqrt{2}\left(\frac{y}{x}\right)} \right| = \log x + C$$

$$\Rightarrow \frac{1}{2\sqrt{2}} \log \left| \frac{x+\sqrt{2}y}{x-\sqrt{2}y} \right| = \log x + C$$

The solution of the given differential equation $\frac{1}{2\sqrt{2}} \log \left| \frac{x+\sqrt{2}y}{x-\sqrt{2}y} \right| = \log x + C$.

6. Show that, differential equation $xdy-ydx=\sqrt{x^2+y^2}dx$ is homogenous and solves it.

Ans: Rewrite the equation in standard form:

$$xdy = \sqrt{x^2 + y^2} dx + ydx$$

$$\Rightarrow \frac{dy}{dx} = \frac{\sqrt{x^2 + y^2} + y}{x}$$

Checking for homogeneity:

$$F(x, y) = \frac{\sqrt{x^2 + y^2} + y}{x}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\sqrt{\lambda^2 x^2 + \lambda^2 y^2} + \lambda y}{\lambda x}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\sqrt{\lambda^2 (x^2 + y^2)} + \lambda y}{\lambda x}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda (\sqrt{x^2 + y^2} + y)}{\lambda (x)}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\sqrt{x^2 + y^2} + y}{x}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y=vx$:

$$\frac{d(vx)}{dx} = \frac{\sqrt{x^2 + (vx)^2} + vx}{x}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = \frac{x(\sqrt{1+v^2} + v)}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = \sqrt{1+v^2} + v$$

$$\Rightarrow x \frac{dv}{dx} = \sqrt{1+v^2}$$

Separate the differentials:

$$\frac{1}{\sqrt{1+v^2}} dv = \frac{dx}{x}$$

Integrate both side:

$$\int \frac{1}{\sqrt{1+v^2}} dv = \int \frac{dx}{x}$$

$$\Rightarrow \log |v + \sqrt{1+v^2}| = \log x + \log C$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow \log \left| \frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}} \right| = \log Cx$$

$$\Rightarrow \log \left| \frac{y + \sqrt{y^2 + x^2}}{x} \right| = \log Cx$$

$$y + \sqrt{y^2 + x^2} = Cx^2$$

7. Show that, differential equation $\left\{ x \cos\left(\frac{y}{x}\right) + y \sin\left(\frac{y}{x}\right) \right\} y dx = \left\{ y \sin\left(\frac{y}{x}\right) - x \cos\left(\frac{y}{x}\right) \right\} x dy$ is homogenous and solves it.

Ans: Rewrite the equation in standard form:

$$\left\{ x \cos\left(\frac{y}{x}\right) + y \sin\left(\frac{y}{x}\right) \right\} y dx = \left\{ y \sin\left(\frac{y}{x}\right) - x \cos\left(\frac{y}{x}\right) \right\} x dy$$

$$\frac{dy}{dx} = \frac{\left\{ x \cos\left(\frac{y}{x}\right) + y \sin\left(\frac{y}{x}\right) \right\} y}{\left\{ y \sin\left(\frac{y}{x}\right) - x \cos\left(\frac{y}{x}\right) \right\} x}$$

Checking for homogeneity:

$$F(x, y) = \frac{\left\{ x \cos\left(\frac{y}{x}\right) + y \sin\left(\frac{y}{x}\right) \right\} y}{\left\{ y \sin\left(\frac{y}{x}\right) - x \cos\left(\frac{y}{x}\right) \right\} x}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\left\{ \lambda x \cos\left(\frac{\lambda y}{\lambda x}\right) + y \sin\left(\frac{\lambda y}{\lambda x}\right) \right\} \lambda y}{\left\{ \lambda y \sin\left(\frac{\lambda y}{\lambda x}\right) - \lambda x \cos\left(\frac{\lambda y}{\lambda x}\right) \right\} \lambda x}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda^2 \left\{ x \cos\left(\frac{y}{x}\right) + y \sin\left(\frac{y}{x}\right) \right\} y}{\lambda^2 \left\{ y \sin\left(\frac{y}{x}\right) - x \cos\left(\frac{y}{x}\right) \right\} x}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\left\{ x \cos\left(\frac{y}{x}\right) + y \sin\left(\frac{y}{x}\right) \right\} y}{\left\{ y \sin\left(\frac{y}{x}\right) - x \cos\left(\frac{y}{x}\right) \right\} x}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y=vx$:

$$\frac{d(vx)}{dx} = \frac{\left\{ x \cos\left(\frac{xv}{x}\right) + vx \sin\left(\frac{vx}{x}\right) \right\} vx}{\left\{ vx \sin\left(\frac{vx}{x}\right) - x \cos\left(\frac{vx}{x}\right) \right\} x}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = \frac{x^2 \{ \cos v + v \sin v \} v}{x^2 \{ v \sin v - \cos v \}}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v \cos v + v^2 \sin v}{v \sin v - \cos v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v \cos v + v^2 \sin v}{v \sin v - \cos v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v \cos v + v^2 \sin v - v^2 \sin v + v \cos v}{v \sin v - \cos v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{2v \cos v}{v \sin v - \cos v}$$

Separate the differentials:

$$\frac{v \sin v - \cos v}{v \cos v} dv = 2 \frac{dx}{x}$$

Integrate both side:

$$\int \frac{v \sin v - \cos v}{v \cos v} dv = 2 \int \frac{dx}{x}$$

$$\Rightarrow \int \frac{v \sin v}{v \cos v} dv - \int \frac{\cos v}{v \cos v} dv = 2 \int \frac{dx}{x}$$

$$\Rightarrow \int \tan v dv - \int \frac{1}{v} dv = 2 \log |x| + \log C$$

$$\Rightarrow \log |\sec v| - \log |v| = \log C |x|^2$$

$$\Rightarrow \log \frac{|\sec v|}{|v|} = \log C |x|^2$$

$$\Rightarrow \sec v = Cvx^2$$

Back substitute $v = \frac{y}{x}$:s

$$\Rightarrow \sec \frac{y}{x} = C \left(\frac{y}{x} \right) x^2$$

$$\Rightarrow \cos \frac{y}{x} = \frac{k}{xy} \quad k = \frac{1}{C}$$

The solution of the given differential equation $\cos \frac{y}{x} = \frac{k}{xy}$.

8. Show that, differential equation $x \frac{dy}{dx} - y + x \sin \left(\frac{y}{x} \right) = 0$ is homogenous and solves it.

Ans: Rewrite the equation in standard form:

$$x \frac{dy}{dx} - y + x \sin \left(\frac{y}{x} \right) = 0$$

$$\frac{dy}{dx} = \frac{y - x \sin \left(\frac{y}{x} \right)}{x}$$

Checking for homogeneity:

$$F(x, y) = \frac{y - x \sin \left(\frac{y}{x} \right)}{x}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda y - \lambda x \sin \left(\frac{\lambda y}{\lambda x} \right)}{\lambda x}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda \left(y - x \sin \left(\frac{y}{x} \right) \right)}{\lambda x}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{y - x \sin \left(\frac{y}{x} \right)}{x}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y = vx$:

$$\frac{d(vx)}{dx} = \frac{vx - x \sin \left(\frac{vx}{x} \right)}{x}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = \frac{x(v - \sin v)}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = v - \sin v$$

$$\Rightarrow x \frac{dv}{dx} = -\sin v$$

Separate the differentials:

$$\frac{1}{\sin v} dv = -\frac{dx}{x}$$

Integrate both side:

$$\int \operatorname{cosec} v dv = -\int \frac{dx}{x}$$

$$\Rightarrow \log |\operatorname{cosec} v - \cot v| = -\log x + \log C$$

$$\Rightarrow \log |\operatorname{cosec} v - \cot v| = \log \frac{C}{x}$$

$$\Rightarrow \operatorname{cosec} v - \cot v = \frac{C}{x}$$

$$\Rightarrow \frac{1}{\sin v} - \frac{\cos v}{\sin v} = \frac{C}{x}$$

$$\Rightarrow 1 - \cos v = \frac{C}{x} \sin v$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow 1 - \cos \frac{y}{x} = \frac{C}{x} \sin \frac{y}{x}$$

$$\Rightarrow x \left(1 - \cos \frac{y}{x} \right) = C \sin \left(\frac{y}{x} \right)$$

The solution of the given differential equation $x \left(1 - \cos \frac{y}{x} \right) = C \sin \left(\frac{y}{x} \right)$.

9. Show that, differential equation $y dx + x \log \left(\frac{y}{x} \right) dy - 2x dy = 0$ is homogenous and solves it.

Ans: Rewrite the equation in standard form:

$$ydx = 2x dy - x \log\left(\frac{y}{x}\right) dy$$

$$\frac{dy}{dx} = \frac{y}{2x - x \log\left(\frac{y}{x}\right)}$$

Checking for homogeneity:

$$F(x, y) = \frac{y}{2x - x \log\left(\frac{y}{x}\right)}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda y}{2\lambda x - \lambda x \log\left(\frac{\lambda y}{\lambda x}\right)}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda y}{\lambda \left(2x - x \log\left(\frac{y}{x}\right)\right)}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{y}{2x - x \log\left(\frac{y}{x}\right)}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y=vx$:

$$\frac{d(vx)}{dx} = \frac{vx}{2x - x \log\left(\frac{vx}{x}\right)}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = \frac{xv}{x(2 - \log(v))}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v}{2 - \log(v)}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v}{2 - \log v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v - 2v + v \log v}{2 - \log v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v \log v - v}{2 - \log v}$$

Separate the differentials:

$$\frac{2 - \log v}{v \log v - v} dv = \frac{dx}{x}$$

Integrate both side:

$$\int \frac{2 - \log v}{v \log v - v} dv = \int \frac{dx}{x}$$

$$\int \frac{1 + 1 - \log v}{v(\log v - v)} dv = \log x + \log C$$

$$\Rightarrow \int \frac{1}{v(\log v - 1)} dv + \int \frac{1 - \log v}{v(\log v - 1)} dv = \log x + \log C$$

$$\Rightarrow \int \frac{1}{v(\log v - 1)} dv - \int \frac{1}{v} dv = \log Cx \quad \dots\dots(1)$$

Solving:

$$\int \frac{1}{v(\log v - 1)} dv$$

Substitute $\log v - 1 = t$:

$$\log v - 1 = t$$

$$\Rightarrow \frac{1}{v} dv = dt$$

Thus, the integral will be:

$$\Rightarrow \int \frac{1}{v(\log v - 1)} dv = \int \frac{dt}{t}$$

$$\Rightarrow \int \frac{1}{v(\log v - 1)} dv = \log t$$

$$\Rightarrow \int \frac{1}{v(\log v - 1)} dv = \log(\log v - 1)$$

Using above result for solving (1):

$$\Rightarrow \log(\log v - 1) - \log v = \log Cx$$

$$\Rightarrow \log \frac{\log v - 1}{v} = \log Cx$$

$$\Rightarrow \frac{\log v - 1}{v} = Cx$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow \frac{\log \frac{y}{x} - 1}{\frac{y}{x}} = Cx$$

$$\Rightarrow \log \frac{y}{x} - 1 = Cx \left(\frac{y}{x} \right)$$

$$\Rightarrow \log \frac{y}{x} - 1 = Cy$$

The solution of the given differential equation $\log \frac{y}{x} - 1 = Cy$.

10. Show that, differential equation $\left(1 + e^{\frac{x}{y}}\right)dx + e^{\frac{x}{y}}\left(1 - \frac{x}{y}\right)dy = 0$ is homogenous and solves it.

Ans: Rewrite the equation in standard form:

$$\left(1 + e^{\frac{x}{y}}\right)dx = -e^{\frac{x}{y}}\left(1 - \frac{x}{y}\right)dy$$

$$\frac{dx}{dy} = -\frac{e^{\frac{x}{y}}\left(1 - \frac{x}{y}\right)}{\left(1 + e^{\frac{x}{y}}\right)}$$

Checking for homogeneity:

$$F(x, y) = -\frac{e^{\frac{x}{y}}\left(1 - \frac{x}{y}\right)}{\left(1 + e^{\frac{x}{y}}\right)}$$

$$\Rightarrow F(\lambda x, \lambda y) = -\frac{e^{\frac{\lambda x}{\lambda y}} \left(1 - \frac{\lambda x}{\lambda y}\right)}{\left(1 + e^{\frac{\lambda x}{\lambda y}}\right)}$$

$$\Rightarrow F(\lambda x, \lambda y) = -\frac{e^{\frac{x}{y}} \left(1 - \frac{x}{y}\right)}{\left(1 + e^{\frac{x}{y}}\right)}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $x=vy$:

$$\frac{d(vy)}{dy} = -\frac{e^{\frac{vy}{y}} \left(1 - \frac{vy}{y}\right)}{1 + e^{\frac{vy}{y}}}$$

$$\Rightarrow v \frac{dy}{dy} + y \frac{dv}{dy} = -\frac{e^v (1-v)}{1 + e^v}$$

$$\Rightarrow v + y \frac{dv}{dy} = -\frac{e^v (1-v)}{1 + e^v}$$

$$\Rightarrow y \frac{dv}{dy} = -\frac{e^v (1-v)}{1 + e^v} - v$$

$$\Rightarrow y \frac{dv}{dy} = \frac{-e^v + ve^v - v - ve^v}{1 + e^v}$$

$$\Rightarrow y \frac{dv}{dy} = \frac{-(e^v + v)}{1 + e^v}$$

Separate the differentials:

$$\frac{1 + e^v}{e^v + v} dv = -\frac{dy}{y}$$

Integrate both side:

$$\int \frac{1 + e^v}{e^v + v} dv = -\int \frac{dy}{y}$$

$$\int \frac{e^v + 1}{e^v + v} dv = -\log y + \log C \quad \dots\dots(1)$$

Solving the LHS integral. Substitute $e^v + v = t$:

$$e^v + v = t$$

$$\Rightarrow (e^v + 1) dv = dt$$

Solving the expression (1):

$$\Rightarrow \int \frac{1}{t} dt = \log \frac{C}{y}$$

$$\Rightarrow \log(t) = \log \frac{C}{y}$$

$$\Rightarrow \log(e^v + v) = \log \frac{C}{y}$$

$$\Rightarrow e^v + v = \frac{C}{y}$$

Back substitute $v = \frac{x}{y}$:

$$\Rightarrow e^{\frac{x}{y}} + \frac{x}{y} = \frac{C}{y}$$

$$\Rightarrow ye^{\frac{x}{y}} + x = C$$

The solution of the given differential equation $ye^{\frac{x}{y}} + x = C$

11. For the differential equation $(x+y)dy + (x-y)dx = 0$. Find the particular solution for the condition $y=1$ when $x=1$.

Ans: Given differential equation is:

$$(x+y)dy + (x-y)dx = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x-y}{x+y}$$

Checking for homogeneity:

$$F(x, y) = -\frac{x-y}{x+y}$$

$$\Rightarrow F(\lambda x, \lambda y) = -\frac{\lambda x - \lambda y}{\lambda x + \lambda y}$$

$$\Rightarrow F(\lambda x, \lambda y) = -\frac{\lambda(x-y)}{\lambda(x+y)}$$

$$\Rightarrow F(\lambda x, \lambda y) = -\frac{x-y}{x+y}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y=vx$:

$$\frac{d(vx)}{dx} = -\frac{x-(vx)}{x+(vx)}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = \frac{x(v-1)}{x(v+1)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v-1}{v+1}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v-1}{v+1} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v-1-v^2-v}{v+1}$$

$$\Rightarrow x \frac{dv}{dx} = -\frac{1+v^2}{v+1}$$

Separate the differentials:

$$\frac{v+1}{1+v^2} dv = -\frac{dx}{x}$$

Integrate both side:

$$\int \frac{v+1}{1+v^2} dv = -\int \frac{dx}{x}$$

$$\Rightarrow \int \frac{v}{1+v^2} dv + \int \frac{1}{1+v^2} dv = -\log x + k$$

$$\Rightarrow \frac{1}{2} \log(1+v^2) + \tan^{-1} v + \log x = k$$

$$\Rightarrow \frac{1}{2} \log[x(1+v^2)] + \tan^{-1} v = k$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow \frac{1}{2} \log \left[x \left(1 + \frac{y^2}{x^2} \right) \right] + \tan^{-1} \frac{y}{x} = k$$

$$\Rightarrow \frac{1}{2} \log \left[\frac{x^2 + y^2}{x} \right] + \tan^{-1} \frac{y}{x} = k$$

Now $y=1$ and $x=1$:

$$\Rightarrow \frac{1}{2} \log \left[\frac{1^2 + 1^2}{1} \right] + \tan^{-1} \frac{1}{1} = k$$

$$k = \frac{1}{2} \log 2 + \frac{\pi}{4}$$

The required particular solution: $\Rightarrow \frac{1}{2} \log \left[\frac{x^2 + y^2}{x} \right] + \tan^{-1} \frac{y}{x} = \frac{1}{2} \log 2 + \frac{\pi}{4}$

12. For the differential equation $x^2 dy + (xy + y^2) dx = 0$. Find the particular solution for the condition $y=1$ when $x=1$.

Ans: Given differential equation is:

$$x^2 dy + (xy + y^2) dx = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{xy + y^2}{x^2}$$

Checking for homogeneity:

$$F(x, y) = -\frac{xy + y^2}{x^2}$$

$$\Rightarrow F(\lambda x, \lambda y) = -\frac{(\lambda x)(\lambda y) + \lambda^2 y^2}{\lambda^2 x^2}$$

$$\Rightarrow F(\lambda x, \lambda y) = -\frac{\lambda^2(xy + y^2)}{\lambda^2 x^2}$$

$$\Rightarrow F(\lambda x, \lambda y) = -\frac{xy + y^2}{x^2}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus it is an homogenous equation.

Let $y=vx$:

$$\frac{d(vx)}{dx} = -\frac{x(vx) + (vx)^2}{x^2}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = -\frac{vx^2 + v^2 x^2}{x^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = -\frac{x^2(v + v^2)}{x^2}$$

$$\Rightarrow v + x \frac{dv}{dx} = -v - v^2$$

$$\Rightarrow x \frac{dv}{dx} = -v^2 - 2v$$

Separate the differentials:

$$\frac{1}{v^2 + 2v} dv = -\frac{dx}{x}$$

Integrate both side:

$$\int \frac{1}{v^2 + 2v} dv = -\int \frac{dx}{x}$$

$$\Rightarrow \frac{1}{2} \int \frac{v+2-v}{v(v+2)} dv = -\log x + \log C$$

$$\Rightarrow \frac{1}{2} \int \frac{v+2}{v(v+2)} dv - \frac{1}{2} \int \frac{v}{v(v+2)} dv = -\log x + \log C$$

$$\Rightarrow \frac{1}{2} \int \frac{1}{v} dv - \frac{1}{2} \int \frac{1}{v+2} dv = -\log x + \log C$$

$$\Rightarrow \frac{1}{2} \log v - \frac{1}{2} \log(v+2) = \log \frac{C}{x}$$

$$\Rightarrow \frac{1}{2} \log \frac{v}{v+2} = \log \frac{C}{x}$$

$$\Rightarrow \frac{v}{v+2} = \left(\frac{C}{x} \right)^2$$

$$\Rightarrow \frac{v}{v+2} = \frac{C^2}{x^2}$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow \frac{\frac{y}{x}}{\frac{y}{x} + 2} = \frac{C^2}{x^2}$$

$$\Rightarrow \frac{x^2 y}{y + 2x} = C^2$$

Now $y=1$ and $x=1$:

$$\Rightarrow \frac{1^2(1)}{1+2(1)} = C^2$$

$$\Rightarrow C^2 = \frac{1}{3}$$

The required particular solution:

$$\Rightarrow \frac{x^2 y}{y + 2x} = \frac{1}{3}$$

$$\Rightarrow y + 2x = 3x^2 y$$

13. For the differential equation $\left[x \sin^2 \left(\frac{x}{y} \right) - y \right] dx + x dy = 0$. Find the particular solution for the condition $y = \frac{\pi}{4}$ when $x=1$.

Ans: Given differential equation is:

$$\left[x \sin^2 \left(\frac{x}{y} \right) - y \right] dx + x dy = 0$$

$$\Rightarrow \frac{dy}{dx} = - \frac{\left[x \sin^2 \left(\frac{x}{y} \right) - y \right]}{x}$$

Checking for homogeneity:

$$F(x, y) = - \frac{x \sin^2 \left(\frac{x}{y} \right) - y}{x}$$

$$\Rightarrow F(\lambda x, \lambda y) = - \frac{\lambda x \sin^2 \left(\frac{\lambda x}{\lambda y} \right) - \lambda y}{\lambda x}$$

$$\Rightarrow F(\lambda x, \lambda y) = - \frac{\lambda \left(x \sin^2 \left(\frac{y}{x} \right) - y \right)}{\lambda x}$$

$$\Rightarrow F(\lambda x, \lambda y) = - \frac{x \sin^2 \left(\frac{y}{x} \right) - y}{x}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y=vx$:

$$\frac{d(vx)}{dx} = - \frac{x \sin^2 \left(\frac{vx}{x} \right) - vx}{x}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = \frac{-x \sin^2(v) + vx}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = -\sin^2 v + v$$

$$\Rightarrow x \frac{dv}{dx} = -\sin^2 v$$

Separate the differentials:

$$\operatorname{cosec}^2 v dv = -\frac{dx}{x}$$

Integrate both side:

$$\int \operatorname{cosec}^2 v dv = -\int \frac{dx}{x}$$

$$\Rightarrow -\cot v = -\log x - \log C$$

$$\Rightarrow \cot v = \log Cx$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow \cot \frac{y}{x} = \log Cx$$

Now $y = \frac{\pi}{4}$ and $x=1$:

$$\cot \frac{\pi}{4} = \log C(1)$$

$$\Rightarrow \log C = \cot \frac{\pi}{4}$$

$$\Rightarrow \log C = 1$$

$$\Rightarrow C = e$$

The required particular solution: $\Rightarrow \cot \frac{y}{x} = \log |ex|$

14. For the differential equation $\frac{dy}{dx} - \frac{y}{x} + \operatorname{cosec}\left(\frac{y}{x}\right) = 0$. Find the particular solution for the condition $y=0$ when $x=1$.

Ans: Given differential equation is:

$$\frac{dy}{dx} - \frac{y}{x} + \operatorname{cosec}\left(\frac{y}{x}\right) = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} - \operatorname{cosec}\left(\frac{y}{x}\right)$$

Checking for homogeneity:

$$F(x, y) = \frac{y}{x} - \operatorname{cosec}\left(\frac{y}{x}\right)$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{\lambda y}{\lambda x} - \operatorname{cosec}\left(\frac{\lambda y}{\lambda x}\right)$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{y}{x} - \operatorname{cosec}\left(\frac{y}{x}\right)$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y=vx$:

$$\frac{d(vx)}{dx} = v - \operatorname{cosec}(v)$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = v - \operatorname{cosec}(v)$$

$$\Rightarrow v + x \frac{dv}{dx} = v - \operatorname{cosec}(v)$$

$$\Rightarrow x \frac{dv}{dx} = -\operatorname{cosec}(v)$$

Separate the differentials:

$$\sin v dv = -\frac{dx}{x}$$

Integrate both side:

$$\int \sin v dv = -\int \frac{dx}{x}$$

$$\Rightarrow -\cos v = -\log|x| - \log C$$

$$\cos v = \log|Cx|$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow \cos \frac{y}{x} = \log|Cx|$$

Now $y=0$ and $x=1$:

$$\Rightarrow \cos \frac{0}{1} = \log|C1|$$

$$\Rightarrow \log C = 1$$

$$\Rightarrow C = e$$

The required particular solution: $\Rightarrow \cos \frac{y}{x} = \log |ex|$

15. For the differential equation $2xy + y^2 - 2x^2 \frac{dy}{dx} = 0$. Find the particular solution for the condition $y=2$ when $x=1$.

Ans: Given differential equation is:

$$2xy + y^2 - 2x^2 \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{2xy + y^2}{2x^2}$$

Checking for homogeneity:

$$F(x, y) = \frac{2xy + y^2}{2x^2}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{2(\lambda x)(\lambda y) + \lambda^2 y^2}{2\lambda^2 x^2}$$

$$\Rightarrow F(\lambda x, \lambda y) = \frac{2xy + y^2}{2x^2}$$

$$\Rightarrow F(x, y) = F(\lambda x, \lambda y)$$

Thus, it is an homogenous equation.

Let $y=vx$:

$$\frac{d(vx)}{dx} = \frac{2x(vx) + (vx)^2}{2x^2}$$

$$\Rightarrow v \frac{dx}{dx} + x \frac{dv}{dx} = \frac{2v + v^2}{2}$$

$$\Rightarrow v + x \frac{dv}{dx} = v + \frac{v^2}{2}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2}{2}$$

Separate the differentials:

$$\frac{dv}{v^2} = \frac{1}{2} \left(\frac{dx}{x} \right)$$

Integrate both side:

$$2 \int \frac{dv}{v^2} = \int \left(\frac{dx}{x} \right)$$

$$\Rightarrow \frac{v^{-2+1}}{-2+1} = \log|x| + C$$

$$-\frac{2}{v} = \log|x| + C$$

Back substitute $v = \frac{y}{x}$:

$$\Rightarrow -\frac{2x}{y} = \log|x| + C$$

Now $y=2$ and $x=1$:

$$\Rightarrow -\frac{2(1)}{2} = \log|1| + C$$

$$\Rightarrow C = -1$$

The required particular solution:

$$\Rightarrow -\frac{2x}{y} = \log|x| - 1.$$

$$\Rightarrow y = \frac{2x}{1 - \log|x|} (x \neq 0, e)$$

16. What substitution should be used for solving homogeneous differential equation

$$\frac{dx}{dy} = h\left(\frac{x}{y}\right).$$

A. $y = vx$

B. $v = yx$

C. $x = vy$

D. $x = v$

Ans: The required substitution will be:

$$\frac{x}{y} = v$$

$$\Rightarrow x = vy$$

The correct answer is (C).

17. Which of the following equation is homogeneous:

A. $(4x+6y+5)dy-(3y+2x+4)dx=0$

B. $(xy)dx-(x^3+y^3)dy=0$

C. $(x^3+2y^2)dx+2xydy=0$

D. $y^2dx+(x^2-xy-y^2)dy=0$

Ans:

For option (A):

$$F(x, y) = \frac{3y + 2x + 4}{4x + 6y + 5}$$

$$F(\lambda x, \lambda y) = \frac{3\lambda y + 2\lambda x + 4}{4\lambda x + 6\lambda y + 5}$$

$$F(\lambda x, \lambda y) \neq F(x, y)$$

For option (B):

$$F(x, y) = \frac{xy}{x^3 + y^3}$$

$$F(\lambda x, \lambda y) = \frac{(\lambda x)(\lambda y)}{(\lambda x)^3 + (\lambda y)^3}$$

$$F(\lambda x, \lambda y) = \frac{xy}{\lambda(x + y)}$$

$$F(\lambda x, \lambda y) \neq F(x, y)$$

For option (C):

$$F(x, y) = -\frac{x^3 + 2y^2}{2xy}$$

$$F(\lambda x, \lambda y) = -\frac{\lambda^3 x^3 + 2\lambda^2 y^2}{2(\lambda x)(\lambda y)}$$

$$F(\lambda x, \lambda y) = -\frac{\lambda x^3 + 2y^2}{2xy}$$

$$F(\lambda x, \lambda y) \neq F(x, y)$$

For option (D):

$$F(x, y) = -\frac{y^2}{x^2 - xy - y^2}$$

$$F(\lambda x, \lambda y) = -\frac{\lambda^2 y^2}{\lambda^2 x^2 - (\lambda x)(\lambda y) - \lambda^2 y^2}$$

$$F(\lambda x, \lambda y) = -\frac{y^2}{x^2 - xy - y^2}$$

$$F(\lambda x, \lambda y) = F(x, y)$$

Thus, the correct answer is option (D).

Exercise 9.5

1. Find the general solution for the differential equation $\frac{dy}{dx} + 2y = \sin x$.

Ans: The given differential equation is: $\frac{dy}{dx} + 2y = \sin x$

It is a linear differential equation of the form $\frac{dy}{dx} + py = Q$, with:

$$p = 2$$

$$Q = \sin x$$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{\int 2 dx}$$

$$\Rightarrow I.F = e^{2x}$$

General solution is of the form:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$ye^{2x} = \int \sin x (e^{2x}) dx + C$$

$$\Rightarrow ye^{2x} = I + C \quad \left(I = \int \sin x (e^{2x}) dx \right) \dots\dots(1)$$

$$I = \int \sin x (e^{2x}) dx$$

$$\Rightarrow I = (\sin x) \int e^{2x} dx - \int ((\sin x)' \int e^{2x} dx) dx$$

$$\Rightarrow I = \frac{e^{2x}}{2} \sin x - \int \left(\cos x \left(\frac{e^{2x}}{2} \right) \right) dx$$

$$\Rightarrow I = \frac{e^{2x}}{2} \sin x - \frac{1}{2} \left[\cos x \int e^{2x} dx - \int ((\cos x)' (\int e^{2x} dx)) dx \right]$$

$$\Rightarrow I = \frac{e^{2x}}{2} \sin x - \frac{1}{2} \left[\frac{e^{2x}}{2} \cos x + \frac{1}{2} \int e^{2x} (\sin x) dx \right]$$

$$\Rightarrow I = \frac{e^{2x}}{2} \sin x - \frac{1}{2} \left[\frac{e^{2x}}{2} \cos x + \frac{1}{2} I \right]$$

$$\Rightarrow I = \frac{e^{2x}}{2} \sin x - \frac{e^{2x}}{4} \cos x - \frac{1}{4} I$$

$$\Rightarrow \frac{5}{4} I = \frac{e^{2x}}{2} \sin x - \frac{e^{2x}}{4} \cos x$$

$$\Rightarrow I = \frac{2e^{2x}}{5} \sin x - \frac{e^{2x}}{5} \cos x$$

$$\Rightarrow I = \frac{e^{2x}}{5} [2 \sin x - \cos x]$$

Back substituting I in expression (1):

$$\Rightarrow ye^{2x} = \frac{e^{2x}}{5} [2 \sin x - \cos x] + C$$

$$\Rightarrow y = \frac{1}{5} (2 \sin x - \cos x) + Ce^{-2x}$$

The general solution for given differential equation is $y = \frac{1}{5} (2 \sin x - \cos x) + Ce^{-2x}$

2. Find the general solution for the differential equation $\frac{dy}{dx} + 3y = e^{-2x}$.

Ans: The given differential equation is: $\frac{dy}{dx} + 3y = e^{-2x}$

It is a linear differential equation of the form $\frac{dy}{dx} + py = Q$, with:

$$p = 3$$

$$Q = e^{-2x}$$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{\int 3 dx}$$

$$\Rightarrow I.F = e^{3x}$$

General solution is of the form:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$\Rightarrow ye^{3x} = \int e^{-2x} (e^{3x}) dx + C$$

$$\Rightarrow ye^{3x} = \int e^{-2x+3x} dx + C$$

$$\Rightarrow ye^{3x} = \int e^x dx + C$$

$$\Rightarrow ye^{3x} = e^x + C$$

$$\Rightarrow y = e^{-2x} + Ce^{-3x}$$

The general solution for given differential equation is $y = e^{-2x} + Ce^{-3x}$.

3. Find the general solution for the differential equation $\frac{dy}{dx} + \frac{y}{x} = x^2$.

Ans: The given differential equation is: $\frac{dy}{dx} + \frac{y}{x} = x^2$

It is a linear differential equation of the form $\frac{dy}{dx} + py = Q$, with:

$$p = \frac{1}{x}$$

$$Q = x^2$$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{\int \frac{1}{x} dx}$$

$$\Rightarrow I.F = e^{\log x}$$

$$\Rightarrow I.F = x$$

General solution is of the form:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$\Rightarrow yx = \int x^2(x) dx + C$$

$$\Rightarrow xy = \int x^3 dx + C$$

$$\Rightarrow xy = \frac{x^{3+1}}{3+1} + C$$

$$\Rightarrow xy = \frac{x^4}{4} + C$$

The general solution for given differential equation is $xy = \frac{x^4}{4} + C$.

4. Find the general solution for the differential equation $\frac{dy}{dx} + (\sec x)y = \tan x \left(0 \leq x \leq \frac{\pi}{2} \right)$.

Ans: The given differential equation is: $\frac{dy}{dx} + (\sec x)y = \tan x$

It is a linear differential equation of the form $\frac{dy}{dx} + py = Q$, with:

$$p = \sec x$$

$$Q = \tan x$$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{\int \sec x dx}$$

$$\Rightarrow I.F = e^{\log(\sec x + \tan x)}$$

$$\Rightarrow I.F = (\sec x + \tan x)$$

General solution is of the form:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$\Rightarrow y(\sec x + \tan x) = \int \tan x (\sec x + \tan x) dx + C$$

$$\Rightarrow y(\sec x + \tan x) = \int \tan x \sec x dx + \int \tan^2 x dx + C$$

$$\Rightarrow y(\sec x + \tan x) = \sec x + \int (\sec^2 x - 1) dx + C$$

$$\Rightarrow y(\sec x + \tan x) = \sec x + \int \sec^2 x dx - \int dx + C$$

$$\Rightarrow y(\sec x + \tan x) = \sec x + \tan x - x + C$$

The general solution for given differential equation is $y(\sec x + \tan x) = \sec x + \tan x - x + C$.

5. Find the general solution for the differential equation $\cos^2 x \frac{dy}{dx} + y = \tan x \left(0 \leq x \leq \frac{\pi}{2} \right)$.

Ans: The given differential equation is: $\cos^2 x \frac{dy}{dx} + y = \tan x$

$$\frac{dy}{dx} + (\sec^2 x) y = \sec^2 x \tan x$$

It is a linear differential equation of the form $\frac{dy}{dx} + py = Q$, with:

$$p = \sec^2 x$$

$$Q = \sec^2 x \tan x$$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{\int \sec^2 x dx}$$

$$\Rightarrow I.F = e^{\tan x}$$

General solution is of the form:

$$y(IF) = \int (Q \times IF) dx + C$$

$$\Rightarrow ye^{\tan x} = \int e^{\tan x} (\sec^2 x \tan x) dx + C$$

$$\Rightarrow ye^{\tan x} = I + C \left(I = \int e^{\tan x} (\sec^2 x \tan x) dx \right) \dots\dots(1)$$

Solving the integral I :

$$I = \int e^{\tan x} (\sec^2 x \tan x) dx$$

Substitute $\tan x = t$: $\tan x = t$

$$\Rightarrow \sec^2 x dx = dt$$

$$\Rightarrow I = \int e^t t dt$$

$$\Rightarrow I = t \int e^t dt - \int \left((t)' \int e^t dt \right) dt$$

$$\Rightarrow I = te^t - \int (e^t) dt$$

$$\Rightarrow I = te^t - e^t$$

Back substitute t :

$$I = \tan x e^{\tan x} - e^{\tan x}$$

Back substitute I in expression (1):

$$\Rightarrow ye^{\tan x} = I + C$$

$$\Rightarrow ye^{\tan x} = \tan x e^{\tan x} - e^{\tan x} + C$$

The general solution for given differential equation is $ye^{\tan x} = \tan x e^{\tan x} - e^{\tan x} + C$.

6. Find the general solution for the differential equation $x \frac{dy}{dx} + 2y = x^2 \log x$.

Ans: The given differential equation is: $x \frac{dy}{dx} + 2y = x^2 \log x$

$$\frac{dy}{dx} + \frac{2}{x}y = x \log x$$

It is a linear differential equation of the form $\frac{dy}{dx} + py = Q$, with:

$$p = \frac{2}{x}$$

$$Q = x \log x$$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{\int \frac{2}{x} dx}$$

$$\Rightarrow I.F = e^{2 \log x}$$

$$\Rightarrow I.F = e^{\log x^2}$$

$$\Rightarrow I.F = x^2$$

General solution is of the form:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$\Rightarrow yx^2 = \int x^2 (x \log x) dx + C$$

$$\Rightarrow yx^2 = \int x^3 \log x dx + C$$

$$\Rightarrow yx^2 = I + C \left(I = \int x^3 \log x dx \right) \dots\dots(1)$$

Solving the integral I :

$$I = \int x^3 \log x dx$$

$$\Rightarrow I = \log x \int x^3 dx - \int ((\log x)' \int x^3 dx) dx$$

$$\Rightarrow I = \frac{x^4}{4} \log x - \int \left(\frac{1}{x} \left(\frac{x^4}{4} \right) \right) dx$$

$$\Rightarrow I = \frac{x^4}{4} \log x - \frac{1}{4} \int x^3 dx$$

$$\Rightarrow I = \frac{x^4}{4} \log x - \frac{1}{4} \left(\frac{x^4}{4} \right)$$

$$\Rightarrow I = \frac{x^4}{4} \log x - \frac{x^4}{16}$$

Back substitute I in expression (1):

$$\Rightarrow yx^2 = I + C$$

$$\Rightarrow yx^2 = \frac{x^4}{4} \log x - \frac{x^4}{16} + C$$

$$\Rightarrow y = \frac{x^2}{16} (4 \log x - 1) + Cx^{-2}$$

The general solution for given differential equation is $y = \frac{x^2}{16} (4 \log x - 1) + Cx^{-2}$

7. Find the general solution for the differential equation $x \log x \frac{dy}{dx} + y = \frac{2}{x} \log x$.

Ans: The given differential equation is: $x \log x \frac{dy}{dx} + y = \frac{2}{x} \log x$

$$\frac{dy}{dx} + \frac{y}{x \log x} = \frac{2}{x^2}$$

It is a linear differential equation of the form $\frac{dy}{dx} + py = Q$, with:

$$p = \frac{1}{x \log x}$$

$$Q = \frac{2}{x^2}$$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{\int \frac{1}{x \log x} dx}$$

Substitute $\log x = t$: $\log x = t$

$$\Rightarrow \frac{1}{x} dx = dt$$

$$\Rightarrow I.F = e^{\int \frac{1}{t} dt}$$

$$\Rightarrow I.F = e^{\log t}$$

$$\Rightarrow I.F = t$$

$$\Rightarrow I.F = \log x$$

General solution is of the form:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$\Rightarrow y \log x = \int \frac{2}{x^2} (\log x) dx + C$$

$$\Rightarrow y \log x = I + C \left(I = \int \frac{2}{x^2} (\log x) dx \right) \quad \dots\dots(1)$$

Solving the integral I:

$$I = \int \frac{2}{x^2} (\log x) dx$$

$$I = 2 \left[\log x \int \frac{1}{x^2} dx - \int \left((\log x)' \int \frac{1}{x^2} dx \right) dx \right]$$

$$I = 2 \left[\log x \left(\frac{-1}{x} \right) - \int \left(\frac{1}{x} \left(\frac{-1}{x} \right) \right) dx \right]$$

$$I = 2 \left[-\frac{\log x}{x} + \int \left(\frac{1}{x^2} \right) dx \right]$$

$$I = 2 \left[-\frac{\log x}{x} - \frac{1}{x} \right]$$

Back substitute I in expression (1):

$$y \log x = I + C$$

$$\Rightarrow y \log x = 2 \left[-\frac{\log x}{x} - \frac{1}{x} \right] + C$$

$$\Rightarrow y \log x = -\frac{2}{x} (\log x + 1) + C$$

The general solution for given differential equation is

$$y \log x = -\frac{2}{x}(\log x + 1) + C$$

8. Find the general solution for the differential equation $(1+x^2)dy + 2xydx = \cot x dx$ ($x \neq 0$).

Ans: The given differential equation is: $(1+x^2)dy + 2xydx = \cot x dx$ ($x \neq 0$)

$$\Rightarrow \frac{dy}{dx} + \frac{2xy}{1+x^2} = \frac{\cot x}{1+x^2}$$

It is a linear differential equation of the form $\frac{dy}{dx} + py = Q$, with:

$$p = \frac{2x}{1+x^2}$$

$$Q = \frac{\cot x}{1+x^2}$$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{\int \frac{2x}{1+x^2} dx}$$

Substitute $\log x = t$: $1+x^2 = t$

$$\Rightarrow 2x dx = dt$$

$$\Rightarrow I.F = e^{\int \frac{1}{t} dt}$$

$$\Rightarrow I.F = e^{\log t}$$

$$\Rightarrow I.F = t$$

$$\Rightarrow I.F = 1 + x^2$$

General solution is of the form:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$\Rightarrow y(1 + x^2) = \int \frac{\cot x}{1 + x^2} (1 + x^2) dx + C$$

$$\Rightarrow y(1 + x^2) = \int \cot x dx + C$$

$$\Rightarrow y(1 + x^2) = \log |\sin x| + C$$

The general solution for given differential equation is $y(1 + x^2) = \log |\sin x| + C$.

9. Find the general solution for the differential equation $x \frac{dy}{dx} + y - x + xy \cot x = 0$ ($x \neq 0$).

Ans: The given differential equation is:

$$x \frac{dy}{dx} + y - x + xy \cot x = 0$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} - 1 + y \cot x = 0$$

$$\Rightarrow \frac{dy}{dx} + \left(\frac{1}{x} + \cot x \right) y = 1$$

It is a linear differential equation of the form $\frac{dy}{dx} + py = Q$, with:

$$p = \left(\frac{1}{x} + \cot x \right)$$

$$Q = 1$$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{\int \left(\frac{1}{x} + \cot x \right) dx}$$

$$\Rightarrow I.F = e^{\int \frac{1}{x} dx + \int \cot x dx}$$

$$\Rightarrow I.F = e^{\log x + \log(\sin x)}$$

$$\Rightarrow I.F = e^{\log(x \sin x)}$$

$$\Rightarrow I.F = x \sin x$$

General solution is of the form:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$\Rightarrow y(x \sin x) = \int (1)(x \sin x) dx + C$$

$$\Rightarrow xy \sin x = I + C \left(I = \int x \sin x dx \right) \dots\dots(1)$$

Solving the integral I :

$$I = \int x \sin x dx$$

$$\Rightarrow I = x \int \sin x dx - \int \left((x)' \int \sin x dx \right) dx$$

$$\Rightarrow I = x(-\cos x) + \int (\cos x) dx$$

$$\Rightarrow I = -x \cos x + \sin x$$

Back substitute I in expression (1):

$$xy \sin x = I + C$$

$$\Rightarrow xy \sin x = -x \cos x + \sin x + C$$

$$\Rightarrow y = -\cot x + \frac{1}{x} + \frac{C}{x \sin x}$$

The general solution for given differential equation is $y = -\cot x + \frac{1}{x} + \frac{C}{x \sin x}$

10. Find the general solution for the differential equation $(x+y) \frac{dy}{dx} = 1$.

Ans: The given differential equation is: $(x+y) \frac{dy}{dx} = 1$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{x+y}$$

$$\Rightarrow \frac{dx}{dy} = x+y$$

$$\Rightarrow \frac{dx}{dy} - x = y$$

It is differential equation of the form $\frac{dx}{dy} + px = Q$, with:

$$p = -1$$

$$Q = y$$

Calculate the integrating factor:

$$I.F = e^{\int p dy}$$

$$\Rightarrow I.F = e^{\int -1 dy}$$

$$\Rightarrow I.F = e^{-y}$$

General solution is of the form:

$$x(I.F) = \int (Q \times I.F) dy + C$$

$$\Rightarrow x(e^{-y}) = \int (y)(e^{-y}) dy + C$$

$$\Rightarrow xe^{-y} = I + C \left(I = \int ye^{-y} dy \right) \dots\dots(1)$$

Solving the integral I :

$$I = \int ye^{-y} dy$$

$$\Rightarrow I = y \int e^{-y} dy - \int ((y)' \int e^{-y} dy) dy$$

$$\Rightarrow I = -ye^{-y} + \int ((1)e^{-y}) dy$$

$$\Rightarrow I = -ye^{-y} + \int e^{-y} dy$$

$$\Rightarrow I = -ye^{-y} - e^{-y}$$

Back substitute I in expression (1):

$$xe^{-y} = I + C$$

$$\Rightarrow xe^{-y} = -ye^{-y} - e^{-y} + C$$

$$\Rightarrow x = -y - 1 + Ce^y$$

$$\Rightarrow x + y + 1 = Ce^y$$

The general solution for given differential equation is $x + y + 1 = Ce^y$

11. Find the general solution for the differential equation $ydx + (x - y^2)dy = 0$.

Ans: The given differential equation is:

$$ydx + (x - y^2)dy = 0$$

$$\Rightarrow \frac{dx}{dy} = \frac{y^2 - x}{y}$$

$$\Rightarrow \frac{dx}{dy} + \left(\frac{1}{y}\right)x = y$$

It is differential equation of the form $\frac{dx}{dy} + px = Q$, with:

$$p = \frac{1}{y}$$

$$Q = y$$

Calculate the integrating factor:

$$I.F = e^{\int p dy}$$

$$\Rightarrow I.F = e^{\int \frac{1}{y} dy}$$

$$\Rightarrow I.F = e^{\log y}$$

$$\Rightarrow I.F = y$$

General solution is of the form:

$$x(I.F) = \int (Q \times I.F) dy + C$$

$$\Rightarrow x(y) = \int (y)(y) dy + C$$

$$\Rightarrow xy = \int y^2 dy + C$$

$$\Rightarrow xy = \frac{y^3}{3} + C$$

$$\Rightarrow x = \frac{y^2}{3} + \frac{C}{y}$$

The general solution for given differential equation is $x = \frac{y^2}{3} + \frac{C}{y}$.

12. Find the general solution for the differential equation $(x+3y^2)\frac{dy}{dx} = y(y_0)$.

Ans: The given differential equation is:

$$(x+3y^2)\frac{dy}{dx} = y$$

$$\Rightarrow \frac{dx}{dy} = \frac{x+3y^2}{y}$$

$$\Rightarrow \frac{dx}{dy} - \left(\frac{1}{y}\right)x = 3y$$

It is differential equation of the form $\frac{dx}{dy} + px = Q$, with:

$$p = -\frac{1}{y}$$

$$Q = 3y$$

Calculate the integrating factor:

$$I.F = e^{\int p dy}$$

$$\Rightarrow I.F = e^{-\int \frac{1}{y} dy}$$

$$\Rightarrow I.F = e^{-\log y}$$

$$\Rightarrow I.F = e^{\log y^{-1}}$$

$$\Rightarrow I.F = \frac{1}{y}$$

General solution is of the form:

$$x(I.F) = \int (Q \times I.F) dy + C$$

$$\Rightarrow x\left(\frac{1}{y}\right) = \int (3y)\left(\frac{1}{y}\right) dy + C$$

$$\Rightarrow \frac{x}{y} = 3 \int dy + C$$

$$\Rightarrow \frac{x}{y} = 3y + C$$

$$\Rightarrow x = 3y^2 + Cy$$

The general solution for given differential equation is $x=3y^2+Cy$.

13. Find particular solution for $\frac{dy}{dx}+2y\tan x=\sin x$ satisfying $y=0$ when $x=\frac{\pi}{3}$.

Ans: The given differential equation is: $\frac{dy}{dx}+2y\tan x=\sin x$

$$\frac{dy}{dx}+(2\tan x)y=\sin x$$

It is differential equation of the form $\frac{dy}{dx}+py=Q$, with:

$$p=2\tan x$$

$$Q=\sin x$$

Calculate the integrating factor:

$$I.F=e^{\int p dx}$$

$$\Rightarrow I.F=e^{2\int \tan x dx}$$

$$\Rightarrow I.F=e^{2\log|\sec x|}$$

$$\Rightarrow I.F=e^{\log(\sec x)^2}$$

$$\Rightarrow I.F=\sec^2 x$$

General solution is of the form:

$$y(I.F)=\int(Q\times I.F)dx+C$$

$$\Rightarrow y \sec^2 x = \int (\sin x)(\sec^2 x) dx + C$$

$$\Rightarrow y \sec^2 x = \int \tan x \sec x dx + C$$

$$\Rightarrow y \sec^2 x = \sec x + C$$

$$\Rightarrow y = \cos x + C \cos^2 x$$

Given $y=0$ when $x=\frac{\pi}{3}$:

$$0 = \cos\left(\frac{\pi}{3}\right) + C \cos^2\left(\frac{\pi}{3}\right)$$

$$\Rightarrow 0 = \frac{1}{2} + C\left(\frac{1}{2}\right)^2$$

$$\Rightarrow C = -2$$

Therefore, the particular solution will be: $\Rightarrow y = \cos x - 2 \cos^2 x$

The particular solution for given differential equation satisfying the given conditions is $y = \cos x - 2 \cos^2 x$.

14. Find particular solution for $(1+x^2)\frac{dy}{dx} + 2xy = \frac{1}{1+x^2}$ satisfying $y=0$ when $x=1$.

Ans: The given differential equation is: $(1+x^2)\frac{dy}{dx} + 2xy = \frac{1}{1+x^2}$

$$\frac{dy}{dx} + \left(\frac{2x}{1+x^2}\right)y = \frac{1}{(1+x^2)^2}$$

It is differential equation of the form $\frac{dy}{dx} + py = Q$, with:

$$p = \frac{2x}{1+x^2}$$

$$Q = \frac{1}{(1+x^2)^2}$$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{\int \frac{2x}{1+x^2} dx}$$

$$\Rightarrow I.F = e^{2 \log |\sec x|}$$

$$\Rightarrow I.F = e^{\log(1+x^2)}$$

$$\Rightarrow I.F = 1+x^2$$

General solution is of the form:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$\Rightarrow y(1+x^2) = \int \left(\frac{1}{(1+x^2)^2} \right) (1+x^2) dx + C$$

$$\Rightarrow y(1+x^2) = \int \frac{1}{1+x^2} dx + C$$

$$\Rightarrow y(1+x^2) = \tan^{-1} x + C$$

Given $y=0$ when $x=1$:

$$0(1+1) = \tan^{-1}(1) + C$$

$$\Rightarrow C + \frac{\pi}{4} = 0$$

$$\Rightarrow C = -\frac{\pi}{4}$$

Therefore, the particular solution will be:

$$\Rightarrow y(1+x^2) = \tan^{-1} x - \frac{\pi}{4}$$

The particular solution for given differential equation satisfying the given conditions is $y(1+x^2) = \tan^{-1} x - \frac{\pi}{4}$.

15. Find particular solution for $\frac{dy}{dx} - 3y \cot x = \sin 2x$ satisfying $y=2$ when $x = \frac{\pi}{2}$.

Ans: The given differential equation is: $\frac{dy}{dx} - 3y \cot x = \sin 2x$

$$\frac{dy}{dx} + (-3 \cot x)y = \sin 2x$$

It is differential equation of the form $\frac{dy}{dx} + py = Q$, with:

$$p = -3 \cot x$$

$$Q = \sin 2x$$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{-3 \int \cot x dx}$$

$$\Rightarrow I.F = e^{-3 \log |\sin x|}$$

$$\Rightarrow I.F = e^{\log \left(\frac{1}{\sin^3 x} \right)}$$

$$\Rightarrow I.F = \frac{1}{\sin^3 x}$$

General solution is of the form:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$\Rightarrow y \left(\frac{1}{\sin^3 x} \right) = \int (\sin 2x) \left(\frac{1}{\sin^3 x} \right) dx + C$$

$$\Rightarrow y \left(\frac{1}{\sin^3 x} \right) = 2 \int (\sin x \cos x) \left(\frac{1}{\sin^3 x} \right) dx + C$$

$$\Rightarrow \frac{y}{\sin^3 x} = 2 \int \left(\frac{\cos x}{\sin^2 x} \right) dx + C$$

$$\Rightarrow \frac{y}{\sin^3 x} = 2 \int \cot x \operatorname{cosec} x dx + C$$

$$\Rightarrow \frac{y}{\sin^3 x} = -2 \operatorname{cosec} x + C$$

$$\Rightarrow y = -2 \sin^2 x + C \sin^3 x$$

Given $y=2$ when $x=\frac{\pi}{2}$:

$$2 = -2 \sin^2 \left(\frac{\pi}{2} \right) + C \sin^3 \left(\frac{\pi}{2} \right)$$

$$\Rightarrow C - 2 = 2$$

$$\Rightarrow C = 4$$

Therefore, the particular solution will be:

$$\Rightarrow y = -2\sin^2 x + 4\sin^3 x$$

The particular solution for given differential equation satisfying the given conditions is $y = -2\sin^2 x + 4\sin^3 x$.

16. Find the equation of a curve passing through the origin given that the slope of the tangent to the curve at any point (x, y) is equal to the sum of the coordinates of the point.

Ans: According to the slope of tangent $\frac{dy}{dx}$ is equal to sum of the coordinates:

$$\frac{dy}{dx} = x + y$$

The given differential equation is: $\frac{dy}{dx} = x + y$

$$\frac{dy}{dx} + (-1)y = x$$

It is differential equation of the form $\frac{dy}{dx} + py = Q$,

with: $p = -1$ and $Q = x$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{-1 \int dx}$$

$$\Rightarrow I.F = e^{-x}$$

General solution is of the form:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$\Rightarrow y(e^{-x}) = \int x(e^{-x}) dx + C$$

$$\Rightarrow ye^{-x} = x \int e^{-x} dx - \int \left((x)' \int e^{-x} dx \right) dx + C$$

$$\Rightarrow ye^{-x} = -xe^{-x} + \int (e^{-x}) dx + C$$

$$\Rightarrow ye^{-x} = -xe^{-x} - e^{-x} + C$$

$$\Rightarrow y = -x - 1 + Ce^x$$

$$\Rightarrow y + x + 1 = Ce^x$$

Given $y=0$ when $x=0$ as it passes through origin:

$$0 + 0 + 1 = Ce^0$$

$$\Rightarrow C = 1$$

Therefore, the equation of the required curve is $y+x+1=e^x$.

17. Find the equation of a curve passing through the point (0,2) given that the sum of the coordinates of any point on the curve exceeds the magnitude of the slope of the tangent to the curve at that point by 5.

Ans: Let the slope of tangent be $\frac{dy}{dx}$.

According to:

$$x + y = \frac{dy}{dx} + 5$$

The given differential equation is: $x + y = \frac{dy}{dx} + 5$

$$\frac{dy}{dx} + (-1)y = x - 5$$

It is differential equation of the form $\frac{dy}{dx} + py = Q$,

with: $p = -1$ and $Q = x - 5$

Calculate the integrating factor:

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{\int -1 dx}$$

$$\Rightarrow I.F = e^{-x}$$

General solution is of the form:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$\Rightarrow y(e^{-x}) = \int (x - 5)(e^{-x}) dx + C$$

$$\Rightarrow y(e^{-x}) = \int x(e^{-x}) dx - 5 \int e^{-x} dx + C$$

$$\Rightarrow ye^{-x} = x \int e^{-x} dx - \int \left((x)' \int e^{-x} dx \right) dx - 5 \int e^{-x} dx + C$$

$$\Rightarrow ye^{-x} = -xe^{-x} + \int (e^{-x})dx + 5e^{-x} + C$$

$$\Rightarrow ye^{-x} = -xe^{-x} - e^{-x} + 5e^{-x} + C$$

$$\Rightarrow ye^{-x} = -xe^{-x} + 4e^{-x} + C$$

$$\Rightarrow y = -x + 4 + Ce^x$$

$$\Rightarrow y + x - 4 = Ce^x$$

Given as it passes through (0,2) :

$$2 + 0 - 4 = Ce^0$$

$$\Rightarrow C = -2$$

Therefore, the equation of the required curve is: $y + x - 4 = -2e^x$

18. Find the integrating factor of the differential equation $x \frac{dy}{dx} - y = 2x^2$.

A. e^{-x}

B. e^{-y}

C. $\frac{1}{x}$

D. x

Ans: Given differential equation is:

$$x \frac{dy}{dx} - y = 2x^2$$

$$\Rightarrow \frac{dy}{dx} - \left(\frac{1}{x}\right)y = 2x^2$$

Thus, it is a linear differential equation of the form $\frac{dy}{dx} + py = Q$:

$$p = -\frac{1}{x}$$

$$I.F = e^{\int p dx}$$

$$\Rightarrow I.F = e^{-\int \frac{1}{x} dx}$$

$$\Rightarrow I.F = e^{-\log|x|}$$

$$\Rightarrow I.F = e^{\log x^{-1}}$$

$$\Rightarrow I.F = \frac{1}{x}$$

Therefore, integrating factor is $\frac{1}{x}$. Thus the correct option is (C).

19. Find the integrating factor of the differential equation $(1-y^2)\frac{dx}{dy} + yx = ay(-1y1)$.

A. $\frac{1}{y^2 - 1}$

B. $\frac{1}{\sqrt{y^2 - 1}}$

C. $\frac{1}{1 - y^2}$

D. $\frac{1}{\sqrt{1-y^2}}$

Ans: Given differential equation is: $(1-y^2)\frac{dx}{dy} + yx = ay$

$$\Rightarrow \frac{dx}{dy} + \left(\frac{y}{1-y^2}\right)x = \frac{ay}{(1-y^2)}$$

Thus, it is a linear differential equation of the form $\frac{dx}{dy} + px = Q$:

$$p = \frac{y}{1-y^2}$$

$$I.F = e^{\int p dy}$$

$$\Rightarrow I.F = e^{\int \frac{y}{1-y^2} dy}$$

$$\Rightarrow I.F = e^{\frac{1}{-2} \int \frac{-2y}{1-y^2} dy}$$

$$\Rightarrow I.F = e^{\frac{1}{-2} \log(1-y^2)}$$

$$\Rightarrow I.F = e^{\log(1-y^2)^{-\frac{1}{2}}}$$

$$\Rightarrow I.F = \frac{1}{\sqrt{1-y^2}}$$

Therefore, integrating factor is $\frac{1}{\sqrt{1-y^2}}$. Thus, the correct option is (D).

Miscellaneous Exercise

1. For each of the differential equations given below, indicate its order and degree (if defined).

(i) $\frac{d^2y}{dx^2} + 5x\left(\frac{dy}{dx}\right)^2 - 6y = \log x$

Ans: The given differential equation is:

$$\frac{d^2y}{dx^2} + 5x\left(\frac{dy}{dx}\right)^2 - 6y - \log x = 0$$

The highest order derivative in the equation is of the term $\frac{d^2y}{dx^2}$, thus the order of the equation is 2 and its highest power is 1. Therefore, its degree is 1.

(ii) $\left(\frac{dy}{dx}\right)^3 - 4\left(\frac{dy}{dx}\right)^2 + 7y = \sin x$

Ans: The given differential equation is:

$$\left(\frac{dy}{dx}\right)^3 - 4\left(\frac{dy}{dx}\right)^2 + 7y - \sin x = 0$$

The highest order derivative in the equation is of the term $\left(\frac{dy}{dx}\right)^3$, thus the order of the equation is 1 and its highest power is 3. Therefore, its degree is 3.

(iii) $\frac{d^4y}{dx^4} - \sin\left(\frac{d^3y}{dx^3}\right) = 0$

Ans: The given differential equation is:

$$\frac{d^4y}{dx^4} - \sin\left(\frac{d^3y}{dx^3}\right) = 0$$

The highest order derivative in the equation is of the term $\frac{d^4y}{dx^4}$, thus the order of the equation is 4.

As the differential equation is not polynomial in its derivative, therefore its degree is not defined.

2. For each of the exercises given below, verify that the given function (implicit or explicit) is a solution of the corresponding differential equation.

(i) $xy = ae^x + be^{-x} + x^2 : x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} - xy + x^2 - 2 = 0$

Ans: The given function is: $xy = ae^x + be^{-x} + x^2$

Take derivative on both side:

$$\Rightarrow y + x \frac{dy}{dx} = ae^x - be^{-x} + 2x$$

Take derivative on both side:

$$\Rightarrow \frac{dy}{dx} + x \frac{d^2y}{dx^2} + \frac{dy}{dx} = ae^x + be^{-x} + 2$$

$$\Rightarrow x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = ae^x + be^{-x} + 2 \quad \dots\dots(1)$$

The given differential equation is: $x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} - xy + x^2 - 2 = 0$

Solving LHS:

Substitute $x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx}$ from the result (1) and xy :

$$\Rightarrow \left(x \frac{d^2y}{dx^2} + 2 \frac{dy}{dx} \right) - xy + x^2 - 2$$

$$\Rightarrow (ae^x + be^{-x} + 2) - (ae^x + be^{-x} + x^2) + x^2 - 2$$

$$\Rightarrow 2 - x^2 + x^2 - 2$$

$$\Rightarrow 0$$

Thus LHS=RHS, the given function is the solution of the given differential equation.

(ii) $y = e^x (a \cos x + b \sin x)$: $\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 2y = 0$

Ans: The given function is: $y = e^x (a \cos x + b \sin x)$

Take derivative on both side:

$$\Rightarrow \frac{dy}{dx} = e^x (a \cos x + b \sin x) + e^x (-a \sin x + b \cos x)$$

$$\Rightarrow \frac{dy}{dx} = e^x ((a+b) \cos x + (b-a) \sin x)$$

Take derivative on both side:

$$\Rightarrow \frac{d^2y}{dx^2} = e^x ((a+b) \cos x + (b-a) \sin x) + e^x (-(a+b) \sin x + (b-a) \cos x)$$

$$\Rightarrow \frac{d^2y}{dx^2} = e^x ((a+b+b-a)\cos x + (b-a-a-b)\sin x)$$

$$\Rightarrow \frac{d^2y}{dx^2} = e^x (2b\cos x - 2a\sin x)$$

The given differential equation is: $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$

Solving LHS:

$$\Rightarrow e^x (2b\cos x - 2a\sin x) - 2e^x ((a+b)\cos x + (b-a)\sin x) + 2y$$

$$\Rightarrow e^x ((2b - 2a - 2b)\cos x + (-2a - 2b + 2a)\sin x) - 2y$$

$$\Rightarrow e^x (-2a\cos x - 2b\sin x) - 2y$$

$$\Rightarrow -2e^x (a\cos x + b\sin x) - 2y$$

$$\Rightarrow 0$$

Thus LHS=RHS, the given function is the solution of the given differential equation.

(iii) $y = x\sin 3x: \frac{d^2y}{dx^2} + 9y - 6\cos 3x = 0$

Ans: The given function is: $y = x\sin 3x$

Take derivative on both side:

$$\Rightarrow \frac{dy}{dx} = \sin 3x + 3x\cos 3x$$

Take derivative on both side:

$$\Rightarrow \frac{d^2y}{dx^2} = 3\cos 3x + 3(\cos 3x + x(-3\sin 3x))$$

$$\Rightarrow \frac{d^2y}{dx^2} = 3\cos 3x + 3\cos 3x - 9x\sin 3x$$

$$\Rightarrow \frac{d^2y}{dx^2} = 6\cos 3x - 9x\sin 3x$$

The given differential equation is: $\frac{d^2y}{dx^2} + 9y - 6\cos 3x = 0$

Solving LHS:

$$\Rightarrow \frac{d^2y}{dx^2} + 9y - 6\cos 3x$$

$$\Rightarrow (6\cos 3x - 9x\sin 3x) + 9(x\sin 3x) - 6\cos 3x$$

$$\Rightarrow 6\cos 3x - 9x\sin 3x + 9x\sin 3x - 6\cos 3x$$

$$\Rightarrow 0$$

Thus LHS=RHS, the given function is the solution of the given differential equation.

(iv) $x^2 = 2y^2 \log y : (x^2 + y^2) \frac{dy}{dx} - xy = 0$

Ans: The given function is: $x^2 = 2y^2 \log y$

Take derivative on both side:

$$\Rightarrow 2x = 2 \left(2y \log y + y^2 \left(\frac{1}{y} \right) \right) \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x}{(2y \log y + y)}$$

Multiply numerator and denominator by y :

$$\Rightarrow \frac{dy}{dx} = \frac{xy}{(2y^2 \log y + y^2)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{xy}{(x^2 + y^2)}$$

The given differential equation is: $(x^2 + y^2) \frac{dy}{dx} - xy = 0$

Solving LHS:

$$\Rightarrow (x^2 + y^2) \frac{dy}{dx} - xy$$

$$\Rightarrow (x^2 + y^2) \left(\frac{xy}{x^2 + y^2} \right) - xy$$

$$\Rightarrow xy - xy$$

$$\Rightarrow 0$$

Thus LHS=RHS, the given function is the solution of the given differential equation.

3. Prove that $x^2 y^2 = c(x^2 + y^2)^2$ is the general solution of differential equation $(x^3 - 3xy^2)dx = (y^3 - 3x^2y)dy$, where c is a parameter.

Ans: Given differential equation:

$$(x^3 - 3xy^2)dx = (y^3 - 3x^2y)dy$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^3 - 3xy^2}{y^3 - 3x^2y}$$

As it can be seen that this is an homogenous equation. Substitute $y=vx$:

$$\Rightarrow \frac{d(vx)}{dx} = \frac{x^3 - 3x(vx)^2}{(vx)^3 - 3x^2(vx)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{x^3(1 - 3v^2)}{x^3(v^3 - 3v)}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1 - 3v^2}{v^3 - 3v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 - 3v^2}{v^3 - 3v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 - 3v^2 - v^4 + 3v^2}{v^3 - 3v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1 - v^4}{v^3 - 3v}$$

Separate the differentials:

$$\frac{v^3 - 3v}{1 - v^4} dv = \frac{dx}{x}$$

Integrate both side:

$$\int \frac{v^3 - 3v}{1 - v^4} dv = \int \frac{dx}{x}$$

$$\Rightarrow \int \frac{v^3 - 3v}{1 - v^4} dv = \log x + \log C$$

$$\Rightarrow I = \log x + \log C \left(I = \int \frac{v^3 - 3v}{1 - v^4} dv \right) \dots\dots(1)$$

Solving integral I:

$$\Rightarrow I = \int \frac{v^3 - 3v}{1 - v^4} dv$$

$$\Rightarrow \frac{v^3 - 3v}{1 - v^4} = \frac{v^3 - 3v}{(1 - v^2)(1 + v^2)}$$

$$\Rightarrow \frac{v^3 - 3v}{1 - v^4} = \frac{v^3 - 3v}{(1 - v)(1 + v)(1 + v^2)}$$

Using partial fraction:

$$\Rightarrow \frac{v^3 - 3v}{1 - v^4} = \frac{A}{1 - v} + \frac{B}{1 + v} + \frac{Cv + D}{1 + v^2}$$

Solving for A,B,C and D:

$$A = -\frac{1}{2}$$

$$B = \frac{1}{2}$$

$$C = -2$$

$$D = 0$$

$$\Rightarrow \frac{v^3 - 3v}{1 - v^4} = \frac{-\frac{1}{2}}{1 - v} + \frac{\frac{1}{2}}{1 + v} + \frac{-2v + 0}{1 + v^2}$$

$$I = -\frac{1}{2} \int \frac{1}{1 - v} dv + \frac{1}{2} \int \frac{1}{1 + v} dv - \int \frac{2v}{1 + v^2} dv$$

$$I = -\frac{1}{2} (-\log(1 - v)) + \frac{1}{2} (\log(1 + v)) - \log(1 + v^2)$$

$$I = \frac{1}{2} (\log(1 - v^2)) - \frac{2}{2} \log(1 + v^2)$$

$$I = \frac{1}{2} \left(\log \frac{(1 - v^2)}{(1 + v^2)^2} \right)$$

$$\Rightarrow I = \frac{1}{2} \left(\log \frac{\left(1 - \frac{y^2}{x^2}\right)}{\left(1 + \frac{y^2}{x^2}\right)^2} \right)$$

$$\Rightarrow I = \frac{1}{2} \left(\log \frac{x^2(x^2 - y^2)}{(x^2 + y^2)^2} \right)$$

$$\Rightarrow I = \frac{1}{2} \left(\log \frac{(x^2 - y^2)}{(x^2 + y^2)^2} \right) + \frac{1}{2} \log x^2$$

$$\Rightarrow I = \frac{1}{2} \left(\log \frac{(x^2 - y^2)}{(x^2 + y^2)^2} \right) + \log x$$

Back substitute I in expression (1):

$$\Rightarrow I = \log x + \log C$$

$$\Rightarrow \frac{1}{2} \left(\log \frac{(x^2 - y^2)}{(x^2 + y^2)^2} \right) + \log x = \log x + \log C$$

$$\Rightarrow \log \frac{(x^2 - y^2)}{(x^2 + y^2)^2} = 2 \log C$$

$$\Rightarrow \frac{x^2 - y^2}{(x^2 + y^2)^2} = C^2$$

$$\Rightarrow x^2 - y^2 = c(x^2 + y^2)^2 \quad (c = C^2)$$

Thus, for given differential equation, its general solution is $x^2 - y^2 = c(x^2 + y^2)^2$.

4. Find the general solution of the differential equation $\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$.

Ans: The given differential equation is:

$$\frac{dy}{dx} + \sqrt{\frac{1-y^2}{1-x^2}} = 0$$

$$\frac{dy}{dx} = -\sqrt{\frac{1-y^2}{1-x^2}}$$

$$\Rightarrow \frac{dy}{\sqrt{1-y^2}} = -\frac{dx}{\sqrt{1-x^2}}$$

Integrate both side:

$$\Rightarrow \int \frac{dy}{\sqrt{1-y^2}} = -\int \frac{dx}{\sqrt{1-x^2}}$$

$$\Rightarrow \sin^{-1} y = -\sin^{-1} x + C$$

$$\Rightarrow \sin^{-1} y + \sin^{-1} x = C$$

Thus, the general solution of given differential equation is $\sin^{-1}y + \sin^{-1}x = C$.

5. Show that the general solution of the differential equation $\frac{dy}{dx} + \frac{y^2+y+1}{x^2+x+1} = 0$ is given by $(x+y+1) = A(1-x-y-2xy)$ where A is a parameter.

Ans: The given differential equation is:

$$\frac{dy}{dx} + \frac{y^2+y+1}{x^2+x+1} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{y^2+y+1}{x^2+x+1}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{y^2 + 2\left(\frac{1}{2}\right)y + \frac{1}{4} - \frac{1}{4} + 1}{x^2 + 2\left(\frac{1}{2}\right)x + \frac{1}{4} - \frac{1}{4} + 1}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{\left(y + \frac{1}{2}\right)^2 + \frac{3}{4}}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}}$$

$$\Rightarrow \frac{dy}{\left(y + \frac{1}{2}\right)^2 + \frac{3}{4}} = -\frac{dx}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}}$$

Integrate both side:

$$\Rightarrow \int \frac{dy}{\left(y + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = -\int \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$$

$$\Rightarrow \frac{1}{\left(\frac{\sqrt{3}}{2}\right)} \tan^{-1} \left[\frac{y + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right] = -\frac{1}{\left(\frac{\sqrt{3}}{2}\right)} \tan^{-1} \left[\frac{x + \frac{1}{2}}{\frac{\sqrt{3}}{2}} \right] + C$$

$$\Rightarrow \frac{2}{\sqrt{3}} \left(\tan^{-1} \left[\frac{2y+1}{\sqrt{3}} \right] + \tan^{-1} \left[\frac{2x+1}{\sqrt{3}} \right] \right) = C$$

$$\Rightarrow \tan^{-1} \left[\frac{2y+1}{\sqrt{3}} \right] + \tan^{-1} \left[\frac{2x+1}{\sqrt{3}} \right] = \frac{\sqrt{3}}{2} C$$

Thus, the general solution for given differential equation is $\tan^{-1} \left[\frac{2y+1}{\sqrt{3}} \right] + \tan^{-1} \left[\frac{2x+1}{\sqrt{3}} \right] = \frac{\sqrt{3}}{2} C$.

6. Find the equation of the curve passing through the point $\left(0, \frac{\pi}{4}\right)$ whose differential equation is $\sin x \cos y dx + \cos x \sin y dy = 0$.

Ans: Given differential equation is:

$$\sin x \cos y dx + \cos x \sin y dy = 0$$

$$\Rightarrow \sin x \cos y dx + \cos x \sin y dy = 0$$

Divide both side by $\cos x \cos y$:

$$\Rightarrow \frac{\sin x \cos y dx + \cos x \sin y dy}{\cos x \cos y} = 0$$

$$\Rightarrow \tan x dx + \tan y dy = 0$$

$$\Rightarrow \tan y dy = -\tan x dx$$

Integrate both side:

$$\Rightarrow \int \tan y dy = -\int \tan x dx$$

$$\Rightarrow \log(\sec y) = -\log(\sec x) + C$$

$$\Rightarrow \log(\sec y) + \log(\sec x) = C$$

$$\Rightarrow \log(\sec x \sec y) = C$$

$$\Rightarrow \sec x \sec y = k \quad (k = e^C)$$

As curve passes through $\left(0, \frac{\pi}{4}\right)$:

$$\sec 0 \sec\left(\frac{\pi}{4}\right) = k$$

$$\Rightarrow k = \sqrt{2}$$

$$\Rightarrow \sec x \sec y = \sqrt{2}$$

Thus, the equation of required curve is $\sec x \sec y = \sqrt{2}$.

7. Find the particular solution of the differential equation $(1+e^{2x})dy+(1+y^2)e^x dx=0$ given that $y=1$ when $x=0$.

Ans: The given differential equation is:

$$(1+e^{2x})dy+(1+y^2)e^x dx=0$$

Divide both side $(1+e^{2x})(1+y^2)$:

$$\frac{dy}{(1+y^2)} + \frac{e^x}{(1+e^{2x})} dx = 0$$

$$\int \frac{dy}{(1+y^2)} = -\int \frac{e^x}{(1+e^{2x})} dx$$

$$\tan^{-1} y = -\int \frac{e^x}{(1+(e^x)^2)} dx$$

Substitute $t = e^x$: $dt = e^x dx$

$$\Rightarrow \tan^{-1} y = -\int \frac{1}{(1+t^2)} dt$$

$$\Rightarrow \tan^{-1} y = -\tan^{-1} t + C$$

$$\Rightarrow \tan^{-1} y = -\tan^{-1} e^x + C$$

$$\Rightarrow \tan^{-1} y + \tan^{-1} e^x = C$$

As $y=1$ when $x=0$:

$$\tan^{-1}(1) + \tan^{-1}(e^0) = C$$

$$\Rightarrow \frac{\pi}{4} + \frac{\pi}{4} = C$$

$$\Rightarrow C = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1} y + \tan^{-1} e^x = \frac{\pi}{2}$$

Thus, the required particular solution is $\tan^{-1} y + \tan^{-1} e^x = \frac{\pi}{2}$.

8. Solve the differential equation $ye^{\frac{x}{y}} dx = \left(xe^{\frac{x}{y}} + y^2 \right) dy$ ($y \neq 0$).

Ans: The given differential equation is:

$$ye^{\frac{x}{y}} dx = \left(xe^{\frac{x}{y}} + y^2 \right) dy$$

$$ye^{\frac{x}{y}} \frac{dx}{dy} = xe^{\frac{x}{y}} + y^2$$

$$\Rightarrow ye^{\frac{x}{y}} \frac{dx}{dy} - xe^{\frac{x}{y}} = y^2$$

$$\Rightarrow e^{\frac{x}{y}} \left[y \frac{dx}{dy} - x \right] = y^2$$

Substitute $z = e^{\frac{x}{y}}$:

$$z = e^{\frac{x}{y}}$$

$$\frac{d}{dy} z = \frac{d}{dy} e^{\frac{x}{y}}$$

$$\Rightarrow \frac{dz}{dy} = \frac{d}{dy} \left(e^{\frac{x}{y}} \right)$$

$$\Rightarrow \frac{dz}{dy} = e^{\frac{x}{y}} \frac{d}{dy} \left(\frac{x}{y} \right)$$

$$\Rightarrow \frac{dz}{dy} = e^{\frac{x}{y}} \left[\left(\frac{1}{y} \right) \frac{dx}{dy} - \frac{x}{y^2} \right]$$

$$\Rightarrow \frac{dz}{dy} = e^{\frac{x}{y}} \left[\frac{y \frac{dx}{dy} - x}{y^2} \right]$$

$$\Rightarrow \frac{dz}{dy} = 1$$

$$\Rightarrow dz = dy$$

$$\Rightarrow \int dz = \int dy$$

$$\Rightarrow z = y + C$$

$$\Rightarrow e^{\frac{x}{y}} = y + C$$

Thus, the required general solution is $e^{\frac{x}{y}} = y + C$.

9. Find a particular solution of the differential equation $(x-y)(dx+dy)=dx-dy$ given that $y=-1$ when $x=0$. Hint (put $x-y=t$).

Ans: Given differential equation is:

$$(x-y)(dx+dy)=dx-dy$$

$$\Rightarrow (x-y)dx - dx = (y-x)dy - dy$$

$$\Rightarrow (x-y+1)dy = (1-x+y)dx$$

$$\Rightarrow \frac{dy}{dx} = \frac{1-x+y}{x-y+1}$$

Put $x-y=t$:

$$x-y=t$$

$$\Rightarrow 1 - \frac{dy}{dx} = \frac{dt}{dx}$$

$$\Rightarrow 1 - \frac{dt}{dx} = \frac{dy}{dx}$$

$$\Rightarrow 1 - \frac{dt}{dx} = \frac{1-t}{1+t}$$

$$\Rightarrow \frac{dt}{dx} = 1 - \frac{1-t}{1+t}$$

$$\Rightarrow \frac{dt}{dx} = \frac{1+t-1+t}{1+t}$$

$$\Rightarrow \frac{dt}{dx} = \frac{2t}{1+t}$$

$$\Rightarrow \frac{1+t}{t} dt = 2dx$$

Integrate both side:

$$\Rightarrow \int \frac{1+t}{t} dt = 2 \int dx$$

$$\Rightarrow \int \frac{1}{t} dt + \int dt = 2x + C$$

$$\Rightarrow \log|t| + t = 2x + C$$

$$\Rightarrow \log|x-y| + x-y = 2x + C$$

$$\Rightarrow \log|x-y| - y = x + C$$

As $y=-1$ when $x=0$:

$$\Rightarrow \log|0-(-1)| - (-1) = 0 + C$$

$$\Rightarrow \log 1 + 1 = C$$

$$\Rightarrow C = 1$$

Thus, the required particular solution is:

$$\log|x-y| - y = x + 1.$$

10. Solve the differential equation $\left[\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}} \right] \frac{dx}{dy} = 1 (x \neq 0).$

Ans: Given differential equation is: $\left[\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}} \right] \frac{dx}{dy} = 1$

$$\Rightarrow \frac{dy}{dx} = \frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{\sqrt{x}} = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}$$

It is linear differential equation of the form $\frac{dy}{dx} + py = Q$:

$$p = \frac{1}{\sqrt{x}}$$

$$Q = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}$$

Calculating integrating factor:

$$I.F = e^{\int p dx}$$

$$I.F = e^{\int \frac{1}{\sqrt{x}} dx}$$

$$I.F = e^{2\sqrt{x}}$$

The general solution is given by:

$$y \times I.F = \int (Q \times I.F) dx + C$$

$$y \times (e^{2\sqrt{x}}) = \int \left(\frac{e^{-2\sqrt{x}}}{\sqrt{x}} \times e^{2\sqrt{x}} \right) dx + C$$

$$\Rightarrow ye^{2\sqrt{x}} = \int \left(\frac{e^{-2\sqrt{x}+2\sqrt{x}}}{\sqrt{x}} \right) dx + C$$

$$\Rightarrow ye^{2\sqrt{x}} = \int \frac{1}{\sqrt{x}} dx + C$$

$$\Rightarrow ye^{2\sqrt{x}} = 2\sqrt{x} + C$$

Thus, the general solution for the given differential equation is $ye^{2\sqrt{x}} = 2\sqrt{x} + C$.

11. Find a particular solution of the differential equation $\frac{dy}{dx} + y \cot x = 4x \operatorname{cosec} x$ ($x \neq 0$) given that $y=0$ when $x = \frac{\pi}{2}$.

Ans: The given differential equation is: $\frac{dy}{dx} + y \cot x = 4x \operatorname{cosec} x$

It is linear differential equation of the form $\frac{dy}{dx} + py = Q$:

$$p = \cot x$$

$$Q = 4x \operatorname{cosec} x$$

Calculating integrating factor:

$$I.F = e^{\int p dx}$$

$$I.F = e^{\int \cot x dx}$$

$$I.F = e^{\log |\sin x|}$$

$$I.F = \sin x$$

The general solution is given by:

$$y \times I.F = \int (Q \times I.F) dx + C$$

$$\Rightarrow y \times \sin x = \int (4x \cos ecx) \sin x dx + C$$

$$\Rightarrow y \sin x = 4 \int x dx + C$$

$$\Rightarrow y \sin x = 4 \left(\frac{x^2}{2} \right) + C$$

$$\Rightarrow y \sin x = 2x^2 + C$$

As $y=0$ when $x=\frac{\pi}{2}$:

$$0 \times \sin \left(\frac{\pi}{2} \right) = 2 \left(\frac{\pi}{2} \right)^2 + C$$

$$C = -2 \left(\frac{\pi^2}{4} \right)$$

$$C = -\frac{\pi^2}{2}$$

Thus, the required particular solution is: $y \sin x = 2x^2 - \frac{\pi^2}{2}$

12. Find a particular solution of the differential equation $(x+1) \frac{dy}{dx} = 2e^{-y} - 1$ given that $y=0$ when $x=0$.

Ans: The given differential equation is: $(x+1) \frac{dy}{dx} = 2e^{-y} - 1$

$$\Rightarrow \frac{dy}{2e^{-y}-1} = \frac{dx}{x+1}$$

Integrate both side:

$$\Rightarrow \int \frac{dy}{2e^{-y}-1} = \int \frac{dx}{x+1}$$

$$\Rightarrow \int \frac{dy}{2e^{-y}-1} = \log(x+1) + \log C \quad \dots\dots(1)$$

Evaluating LHS integral:

$$\Rightarrow \int \frac{dy}{2e^{-y}-1} = \int \frac{e^y dy}{2-e^y}$$

Put $t=2-e^y$:

$$t = 2 - e^y$$

$$dt = -e^y dy$$

$$\Rightarrow \int \frac{dy}{2e^{-y}-1} = -\int \frac{dt}{t}$$

$$\Rightarrow \int \frac{dy}{2e^{-y}-1} = -\log(t)$$

$$\Rightarrow \int \frac{dy}{2e^{-y}-1} = \log \frac{1}{t}$$

$$\Rightarrow \int \frac{dy}{2e^{-y}-1} = \log \frac{1}{2-e^y}$$

Back substituting in expression (1):

$$\Rightarrow \int \frac{dy}{2e^{-y}-1} = \log(x+1) + \log C$$

$$\Rightarrow \log\left(\frac{1}{2-e^y}\right) = \log C(x+1)$$

$$\Rightarrow 2 - e^y = \frac{1}{C(x+1)}$$

As $y=0$ when $x=0$:

$$\Rightarrow 2 - e^0 = \frac{1}{C(0+1)}$$

$$\Rightarrow 2 - 1 = \frac{1}{C}$$

$$\Rightarrow C = 1$$

Thus, the required particular solution is:

$$\Rightarrow 2 - e^y = \frac{1}{(x+1)}$$

$$\Rightarrow e^y = 2 - \frac{1}{(x+1)}$$

$$\Rightarrow e^y = \frac{2x+2-1}{(x+1)}$$

$$\Rightarrow e^y = \frac{2x+1}{x+1}$$

$$\Rightarrow y = \log\left(\frac{2x+1}{x+1}\right)$$

Thus, for given conditions the particular solution is $y = \log\left(\frac{2x+1}{x+1}\right)$.

13. The general solution of the differential equation $\frac{ydx - xdy}{y} = 0$.

(A) $xy = C$

(B) $x = Cy^2$

(C) $y = Cx$

(D) $y = Cx^2$

Ans: Given differential equation:

$$\frac{ydx - xdy}{y} = 0$$

Divide both side by x :

$$\Rightarrow \frac{ydx - xdy}{xy} = 0$$

$$\Rightarrow \frac{dx}{x} - \frac{dy}{y} = 0$$

Integrate both side:

$$\Rightarrow \int \frac{dx}{x} - \int \frac{dy}{y} = 0$$

$$\Rightarrow \log|x| - \log|y| = \log k$$

$$\Rightarrow \log\left|\frac{x}{y}\right| = \log k$$

$$\Rightarrow \frac{x}{y} = k$$

$$\Rightarrow y = Cx \left(C = \frac{1}{k} \right)$$

Thus, the correct option is (C)

14. Find the general solution of a differential equation of the type $\frac{dx}{dy} + P_1x = Q_1$.

(A) $ye^{\int P_1 dy} = \int \left(Q_1 e^{\int P_1 dy} \right) dy + C$

(B) $ye^{\int P_1 dx} = \int \left(Q_1 e^{\int P_1 dx} \right) dy + C$

(C) $xe^{\int P_1 dy} = \int \left(Q_1 e^{\int P_1 dy} \right) dy + C$

(D) $xe^{\int P_1 dy} = \int \left(Q_1 e^{\int P_1 dx} \right) dy + C$

Ans: The given differential equation is: $\frac{dx}{dy} + P_1x = Q_1$

It is a linear differential equation and its general solution is:

$$xe^{\int P_1 dy} = \int \left(Q_1 e^{\int P_1 dy} \right) dy + C$$

With integrating factor $I.F = e^{\int P_1 dy}$.

Thus, the correct option is (C).

15. Find the general solution of the differential equation $e^x dy + (ye^x + 2x) dx = 0$.

(A) $xe^y + x^2 = C$

(B) $xe^y + y^2 = C$

(C) $ye^x + x^2 = C$

(D) $ye^y + x^2 = C$

Ans: The given differential equation is: $e^x dy + (ye^x + 2x) dx = 0$

$$\Rightarrow e^x \frac{dy}{dx} + ye^x = -2x$$

$$\Rightarrow \frac{dy}{dx} + y = -2xe^{-x}$$

The given differential equation is of the form: $\frac{dy}{dx} + Py = Q$

$$\Rightarrow P = 1$$

$$\Rightarrow Q = -2xe^{-x}$$

Calculating integrating factor:

$$I.F = e^{\int P dx}$$

$$\Rightarrow I.F = e^{\int dx}$$

$$\Rightarrow I.F = e^x$$

It is a linear differential equation and its general solution is:

$$y(I.F) = \int (Q \times I.F) dx + C$$

$$y(e^x) = \int (-2xe^{-x} \times e^x) dy + C$$

$$\Rightarrow ye^x = -2 \int x dx + C$$

$$\Rightarrow ye^x = -2 \left(\frac{x^2}{2} \right) + C$$

$$\Rightarrow ye^x + x^2 = C$$

Thus, the correct answer is option (C).